BT1010 Introduction to Life Sciences

Lecture 4: Cell Organelles and Nuclear Architecture
03/11/2025



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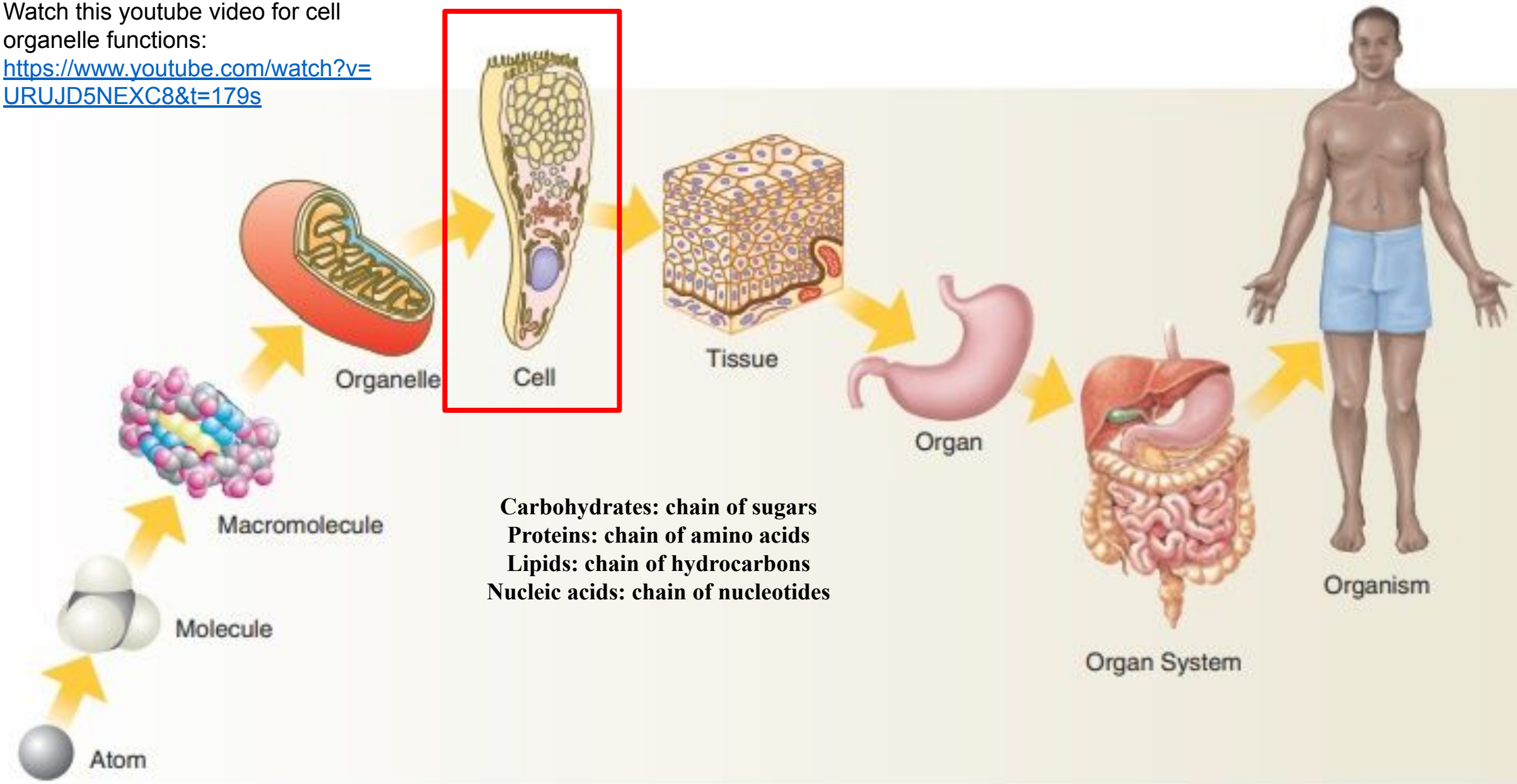
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Organization level of the body

Watch this youtube video for cell organelle functions: https://www.youtube.com/watch?v=URUJD5NEXC8&t=179s

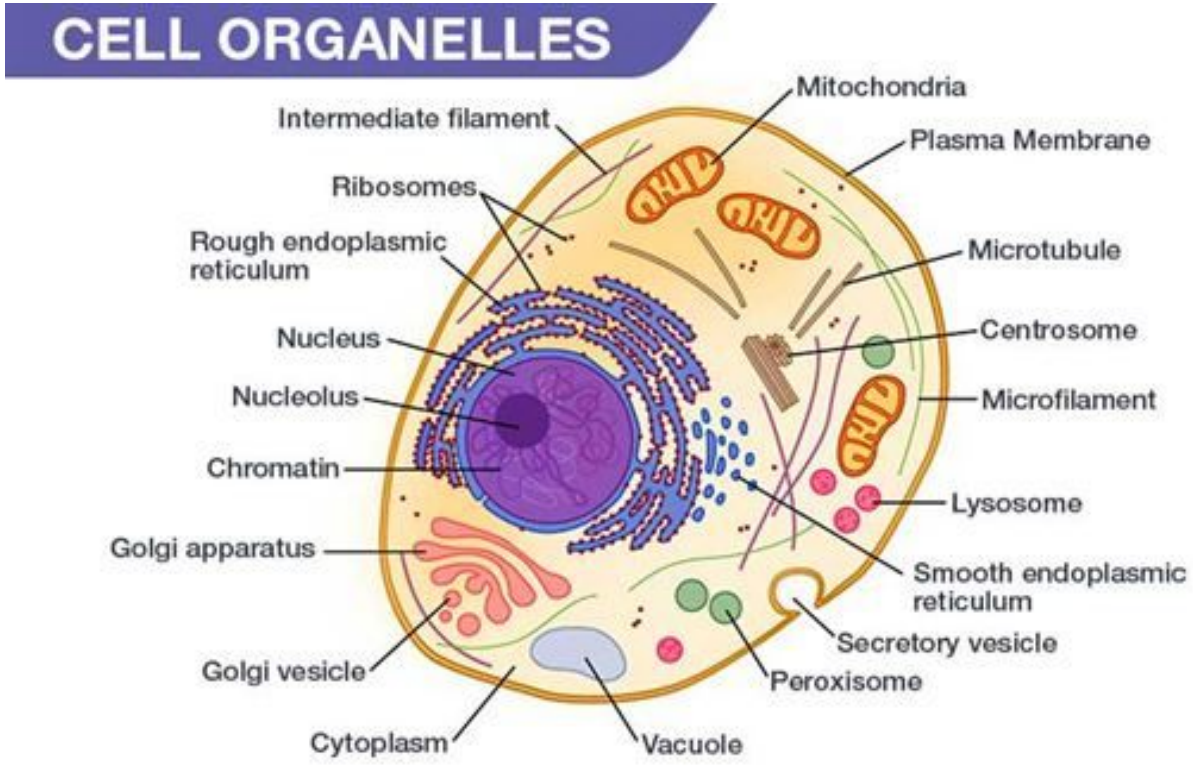
Basic unit of life



Carbohydrates: chain of sugars
Proteins: chain of amino acids
Lipids: chain of hydrocarbons
Nucleic acids: chain of nucleotides

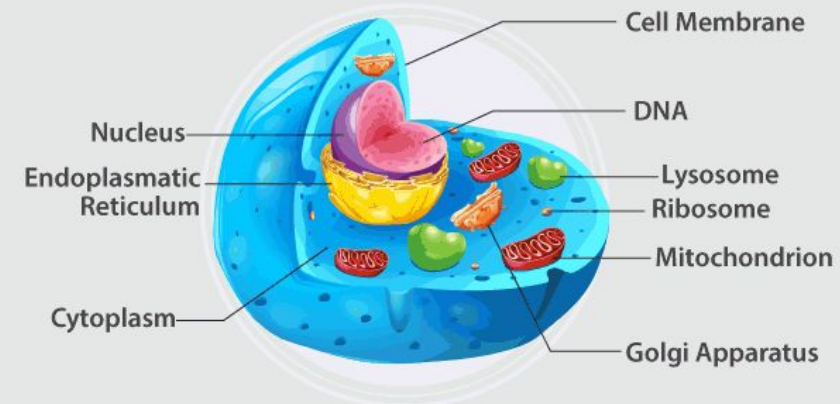
Cell Organelles

- Cell organelles are specialized entities present inside a particular type of cell that performs a specific function.
- They are involved in many processes, for example energy production, building proteins and secretions, destroying toxins, and responding to external signals.
- Organelles are considered either membranous or non-membranous.
- **Membranous organelles** possess their own plasma membrane to create a lumen separate from the cytoplasm. Endoplasmic reticulum, nucleus, golgi apparatus, mitochondria.
- **Non-membranous organelles** are not surrounded by a plasma membrane. Most non-membranous organelles are part of the cytoskeleton, the major support structure of the cell. These include: ribosomes, filaments, microtubules, and centrioles.

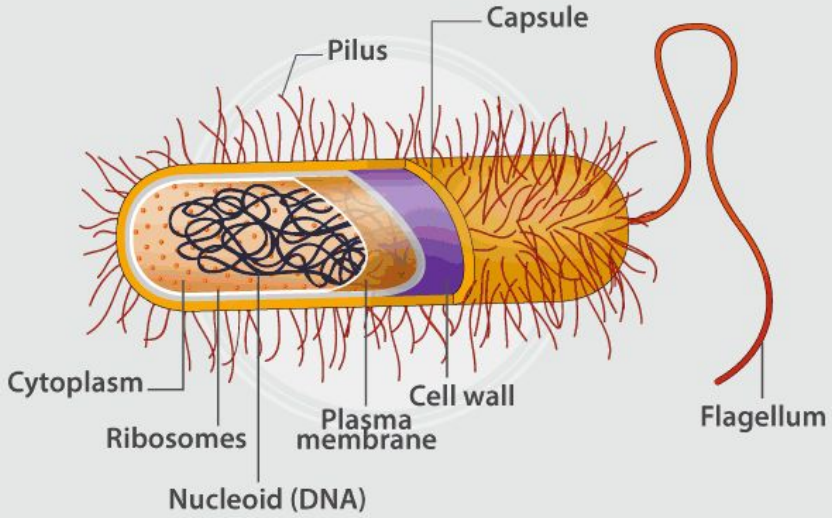


Organelle	*Responsible for*
Nucleus	Housing DNA, 'brain' of the cell
Mitochondria	Energy production, 'power house' of the cell
Golgi apparatus	Sorting, packaging and transport of proteins
Endoplasmic reticulum	Synthesis and processing of proteins, lipid expression
Chloroplast	Photosynthesis, only present in plants
Flagellum	Locomotion and sensory functions
Vacuole	Storage and maintaining homeostasis
Lysosome	Digestions of larger molecules
Peroxisome	Degradation of hydrogen peroxide
Ribosome	Synthesis of proteins
Proteasome	Break down of proteins with expired function

Prokaryotic and Eukaryotic cells



Eukaryotic Cell



Prokaryotic Cell

	Prokaryotes	Eukaryotes
Type of Cell	Always unicellular	Unicellular and multi-cellular
Cell size	0.2 µm – 2.0 µm in diameter	10 µm – 100 µm in diameter
Cell wall	Present	Present/Absent
Nucleus	Absent	Present
Ribosomes	Present. Smaller in size and spherical in shape	Present. Comparatively larger in size
DNA	Circular	Linear
Mitochondria	Absent	Present
Cytoplasm	Present, but cell organelles absent	Present, cell organelles present
Endoplasmic reticulum	Absent	Present
Plasmids	Present	Very rarely found in eukaryotes
Ribosome	Small ribosomes	Large ribosomes
Lysosome	Lysosomes and centrosomes are absent	Lysosomes and centrosomes are present
Cell division	Through binary fission	Through mitosis
Flagella	The flagella are smaller in size	The flagella are larger in size
Reproduction	Asexual	Both asexual and sexual
Example	Bacteria and Archaea	Plant and Animal cell

Difference between plant cells and animal cells (both are eukaryotic cells)

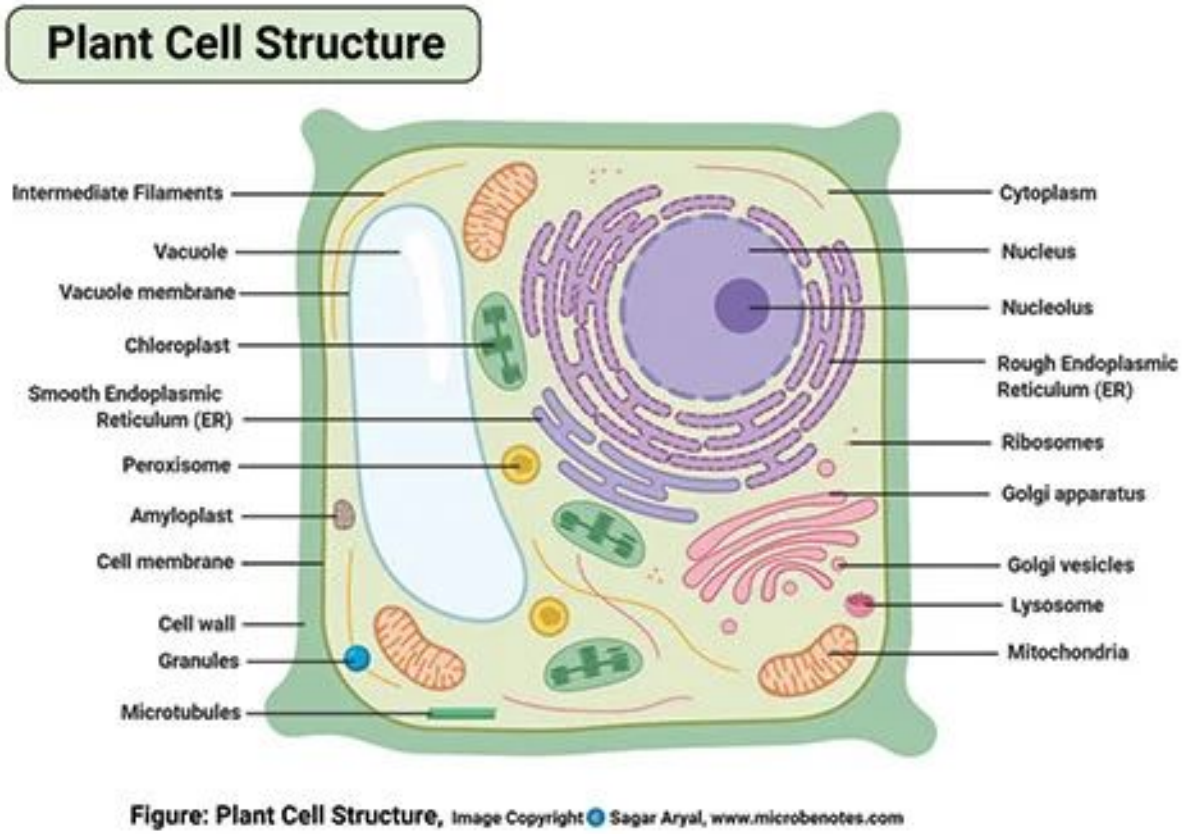


Figure: Plant Cell Structure, Image Copyright © Sagar Aryal, www.microbenotes.com

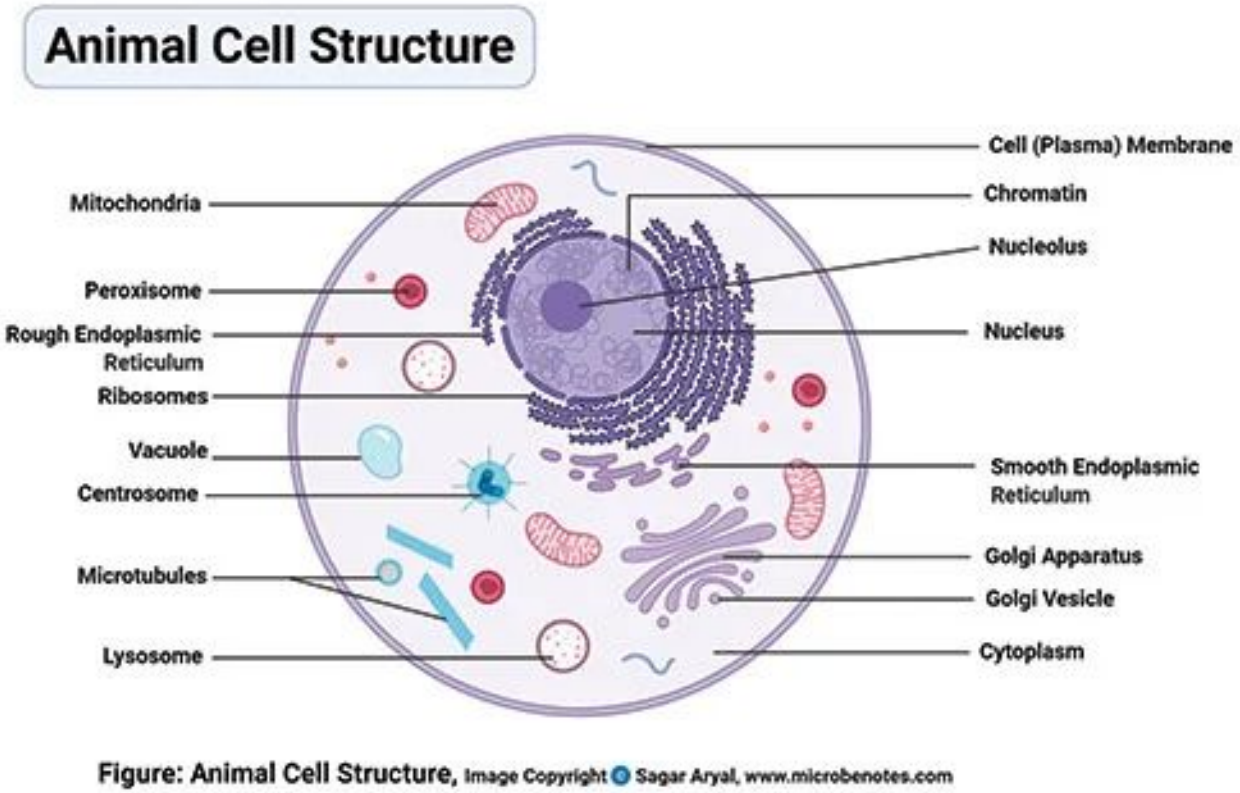


Figure: Animal Cell Structure, Image Copyright © Sagar Aryal, www.microbenotes.com

- Plant cells have a cell wall, but animals cells do not. Cell walls provide support and give shape to plants.
- Plant cells have chloroplasts, but animal cells do not. Chloroplasts enable plants to perform photosynthesis to make food.
- Plant cells usually have one or more large vacuole(s), while animal cells have smaller vacuoles, if any are present. Large vacuoles help provide shape and allow the plant to store water and food for future use.
- Plant cells do not have centrosomes, but they have MTOC (microtubule organizing center) to coordinate their microtubules.

https://microbenotes.com/cell-organelles/

Nucleus (Director/Brain of the cell)

- Only present in Eukaryotic cells.
- Covered by a double membranous nuclear membrane.
- Contains DNA, RNA, Protein, nucleolus, and Chromatin network.
- Controls the activity of the cell. Starts cell division. It contains the genetic material of the cell which is responsible for hereditary characters.
- Every cell, except gametes, in a human body has 2 meters long DNA packaged into the cell nucleus.
- Nuclear pore regulates the movement of RNA and proteins in and out of the nucleus.
- Nucleolus is a non-membrane bound organelle that manufactures ribosomes.

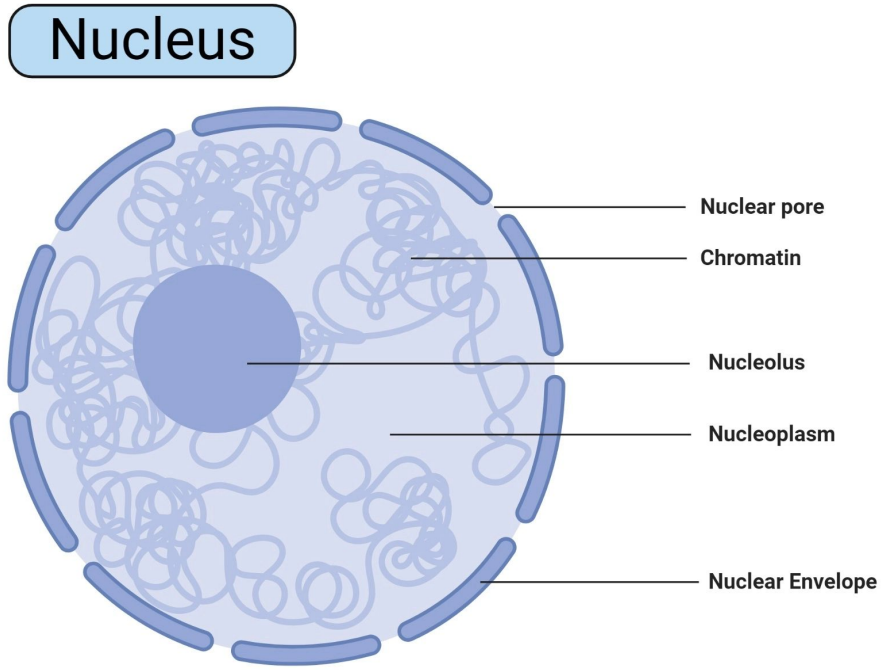
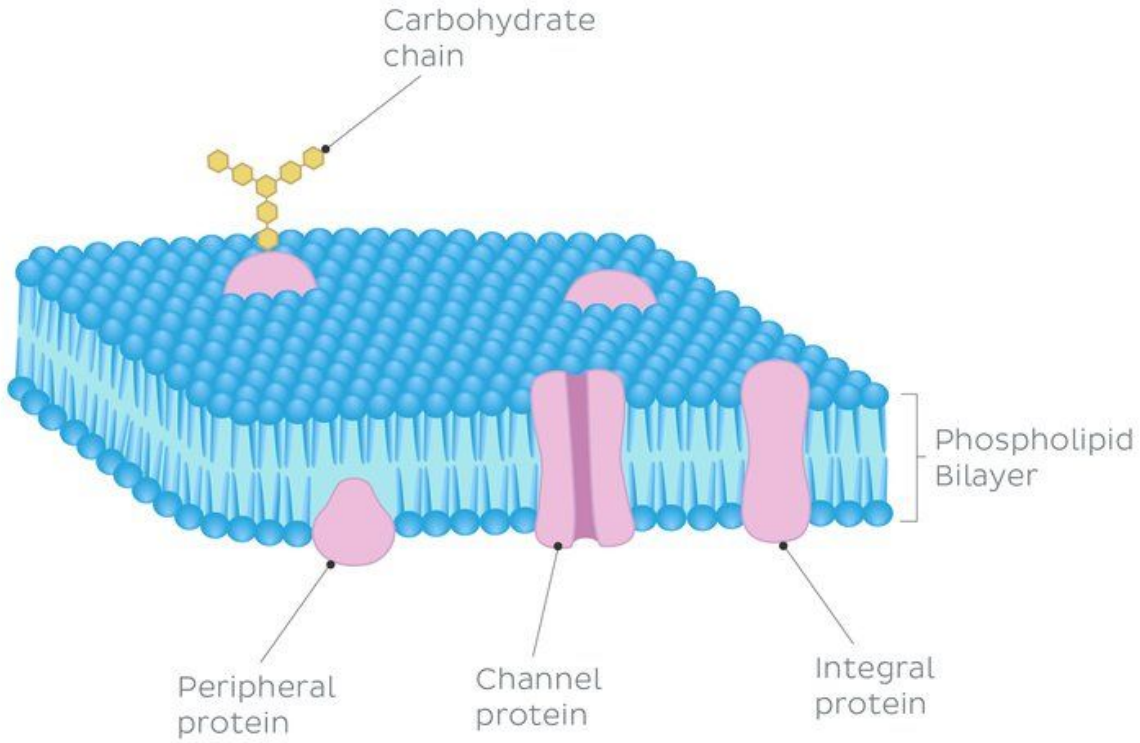


Figure: Nucleus, Image Copyright © Sagar Aryal, www.microbenotes.com

https://youtu.be/HxdajtjxRvg

Plasma Membrane

- The plasma membrane surrounds the cell to create a barrier between the cytosol and the extracellular matrix.
- The structure of the membrane resembles a **fluid mosaic** made up of phospholipids, cholesterol, and membrane proteins.
- **Phospholipid molecules**, the main structural components of the membrane, form an amphipathic bilayer. An **amphipathic** structure is both hydrophilic and hydrophobic; part of the structure has a high affinity for water and the other part is repelled by water. The inner surface of each layer is made up of lipid chains and thus is **hydrophobic**. The outer surface of each layer is made up of the polar heads of the phospholipids and is **hydrophilic**.

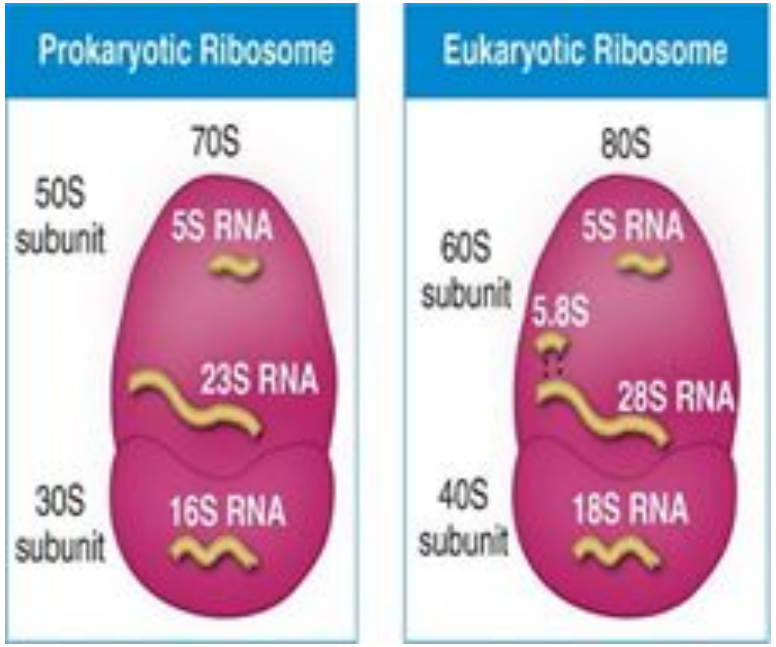


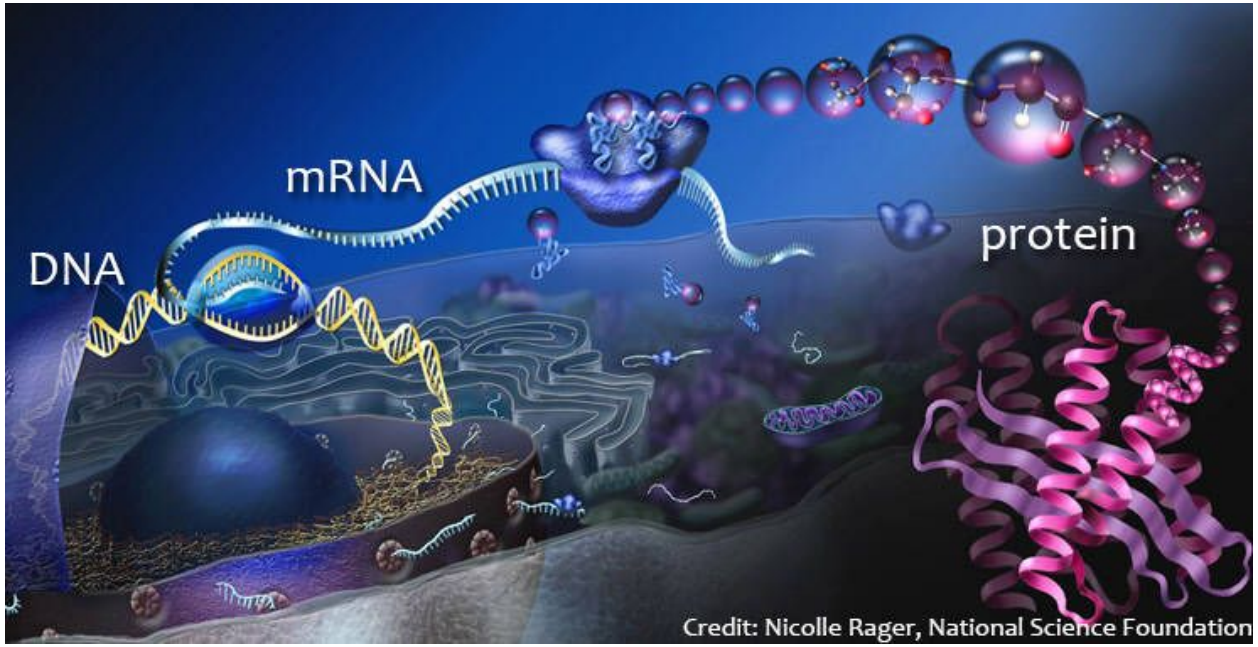
Proteins associated with the plasma membrane are either peripheral membrane proteins or integral membrane proteins. **Peripheral membrane proteins** interact closely with the membrane through ionic interactions. **Integral membrane proteins** are embedded within or pass through the lipid bilayer. There are six broad categories of integral membrane proteins: pumps, channels, receptors, linkers, enzymes, and structural proteins.

- **Pumps** - transport ions, sugars, and amino acids across membranes
- **Channels** - allow small ions and molecules to pass freely in and out of the cell
- **Receptors** - recognise and bind with ligands
- **Linkers** - anchor the cytoskeleton to the extracellular matrix
- **Enzymes** - have many roles, for example ATPases participate in ion pumping
- **Structural proteins** - form junctions with neighboring cells

Ribosomes (Protein factories)

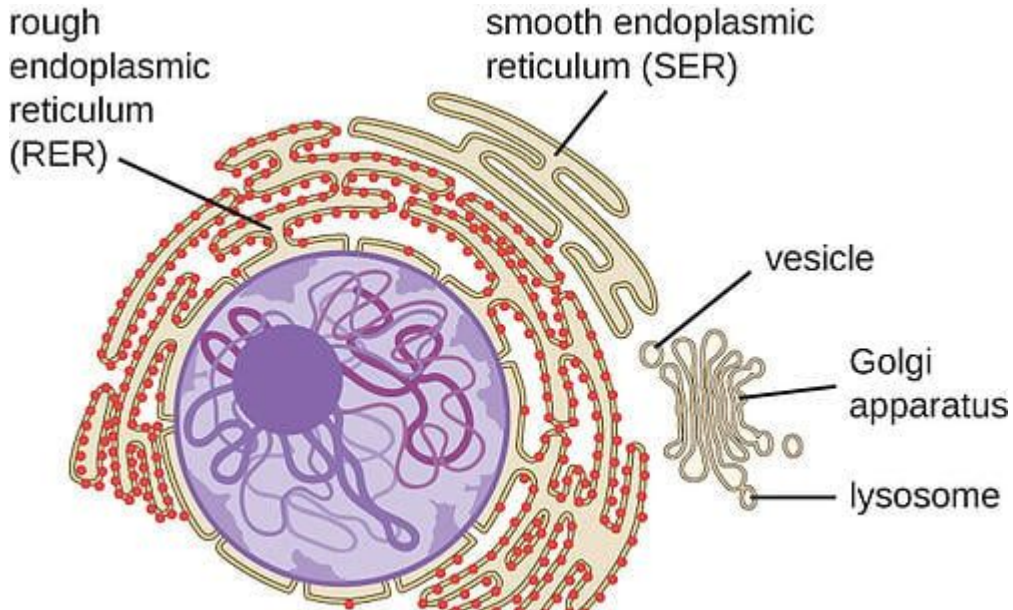
- Ribosomes are non-membranous organelles. They are made up of proteins and RNA.
- Ribosomes, either free in the cytosol or associated with rER, **synthesize proteins** as polypeptide chains. This occurs through the translation of RNA.
- Specifically, ribosomes bind to messenger RNA, abbreviated mRNA. The ribosome reads a series of nucleotide bases in groups of three called **codons**. The first codon read is the start codon. Each codon following the start codon represents a specific amino acid that is then brought to the ribosome by **transfer RNA**, abbreviated tRNA.
- The bond between two amino acids in a polypeptide chain is referred to as a **peptide bond**.





Endoplasmic Reticulum (ER)

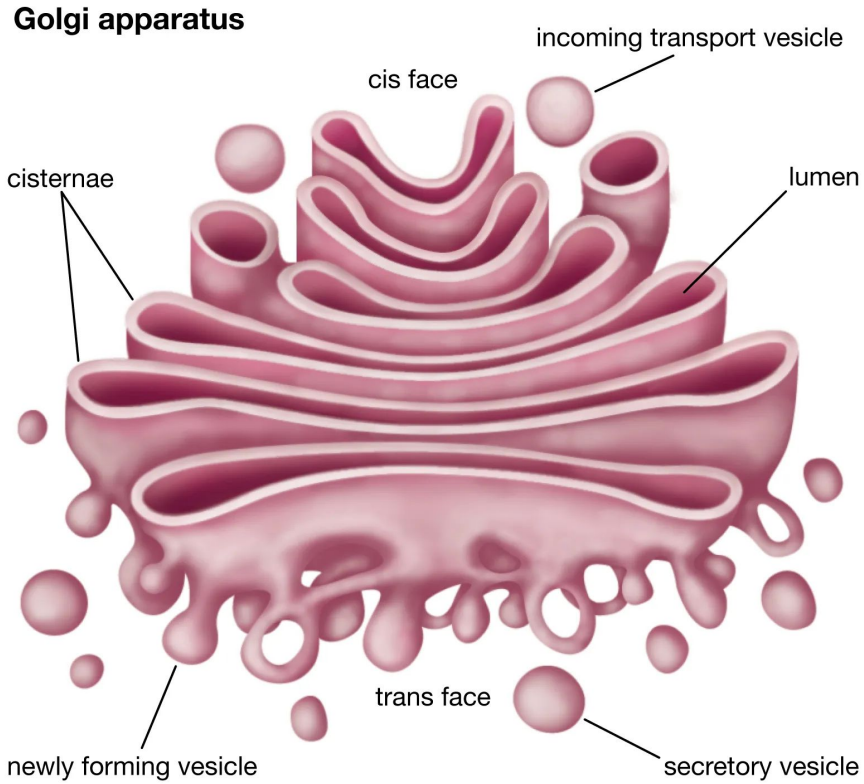
- The endoplasmic reticulum (ER) is a large network of membranes responsible for the production of proteins, metabolism and transportation of lipids, and detoxification of poisons.
- There are two types of endoplasmic reticulum with separate functions: **smooth** endoplasmic reticulum and **rough** endoplasmic reticulum. The presence or absence of ribosomes in the ER's plasma membrane determines whether it is classified as smooth or rough ER.
- The outer plasma membrane of **rough endoplasmic reticulum (rER)** is carpeted with ribosomes, causing it to appear dotted under a microscope. Protein production occurs in the ribosomes of rER. The ribosomes synthesize a peptide strand which enters the lumen of the rER and folds into its functional shape. From there it will be transported to the Golgi apparatus in a membrane bound vesicle formed from budding of the rER membrane.
- **Smooth endoplasmic reticulum**, abbreviated sER, lacks ribosomes and thus appears smooth under a microscope. Its functions vary among cell types. For example, sER in cells of the liver have detoxifying functions while sER in cells of the endocrine system mainly produce steroid hormones. Detoxification occurs through enzymes associated with the sER membrane and usually involves adding hydroxyl groups to molecules. The presence of hydroxyl groups makes the molecules more water soluble and therefore able to be flushed from the body through the urinary tract.



Golgi Apparatus (Shipping department of the cell)

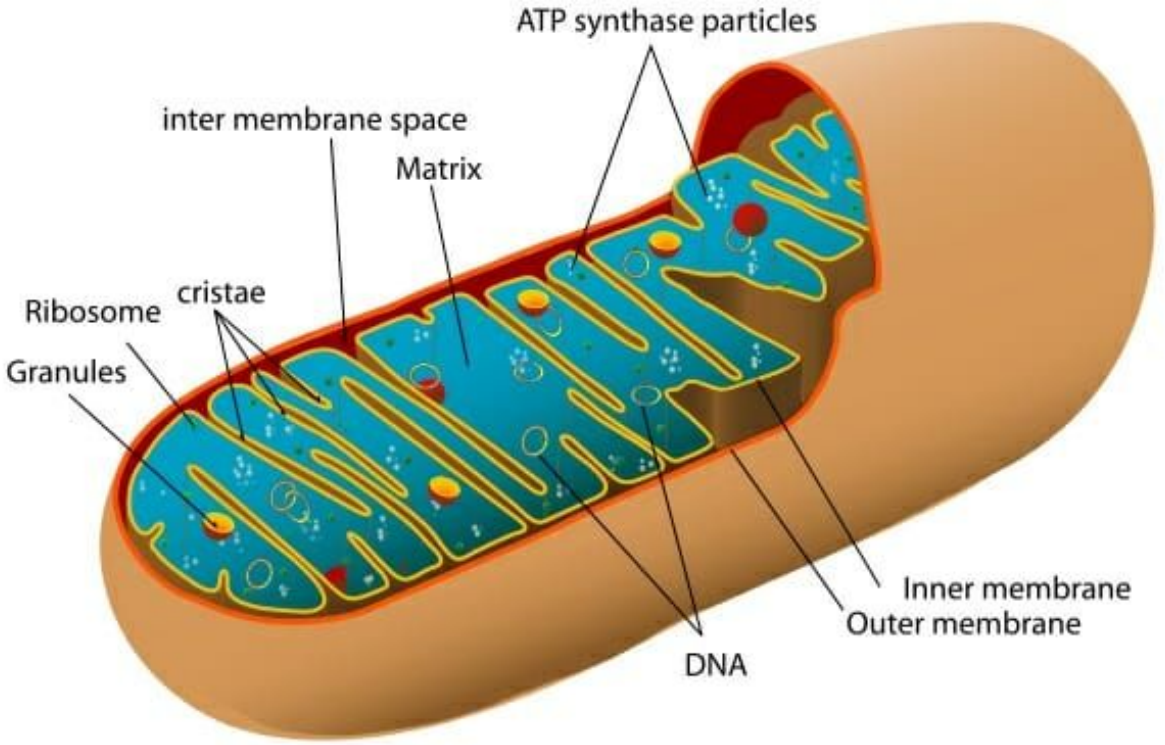
- The Golgi apparatus appears as a series of flattened, membranous sacs, or **cisternae**, that resemble a stack of pancakes just off the rough endoplasmic reticulum. It receives vesicles containing proteins recently produced by the rER. The Golgi apparatus can be compared to a warehouse or post office for newly formed proteins. Here the proteins are further modified, packaged, and sent off to their final destinations in the cell or body.

- Because the Golgi apparatus receives and sends off vesicles from opposite sides of its stack of cisternae, it is considered polar, meaning it has a directional structure. The **cis-face** is located near the rER and receives vesicles. The **trans-face** is on the opposite side of the organelle and releases vesicles through budding of the plasma membrane.



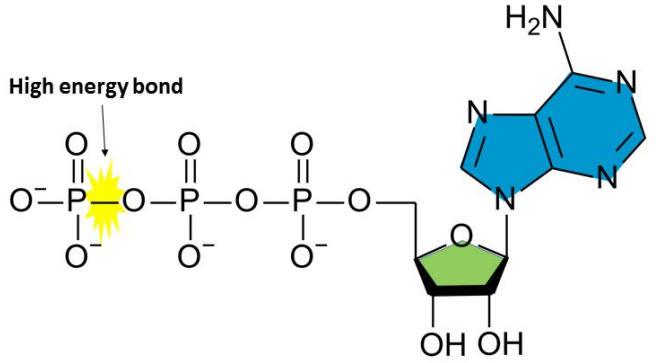
Mitochondria: the powerhouse of the cell

- The mitochondrion, mitochondria denotes plural, is a double membrane bound organelle. Its inner membrane contains many infoldings called **cristae**. The space between the outer and inner membranes is referred to as the **intermembrane space** and the **matrix** is the space inside the inner membrane. Free ribosomes and mitochondrial DNA can be found in the matrix. **Mitochondrial DNA is unique in that it is entirely maternally inherited.**
- Mitochondria are the powerhouses of the cell. **Cellular respiration**, the generation of energy from sugars and fats, occurs in these organelles. Some of the enzymes that catalyze respiration are found within the matrix. Other proteins involved in these reactions are built into the wall of the inner membrane. The cristae of the inner membrane are highly convoluted to increase surface area. This allows for more proteins lining the membrane and thus greater productivity.



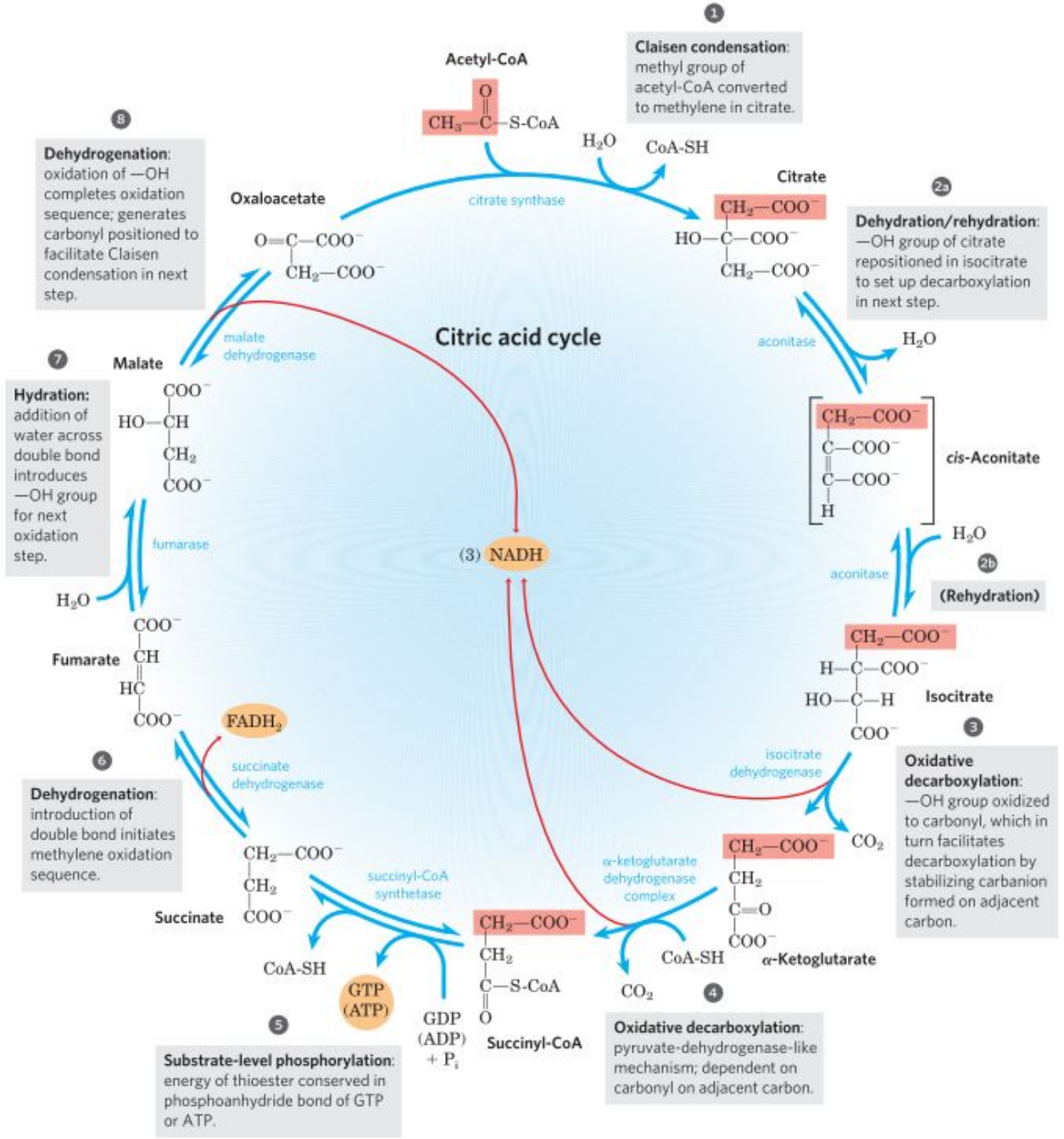
Cellular energy can not be provided in the form of a current. It is provided in the form of an ATP.

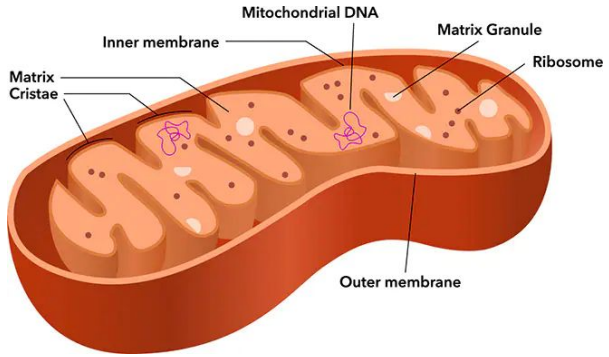
Adenosine Triphosphate (ATP)





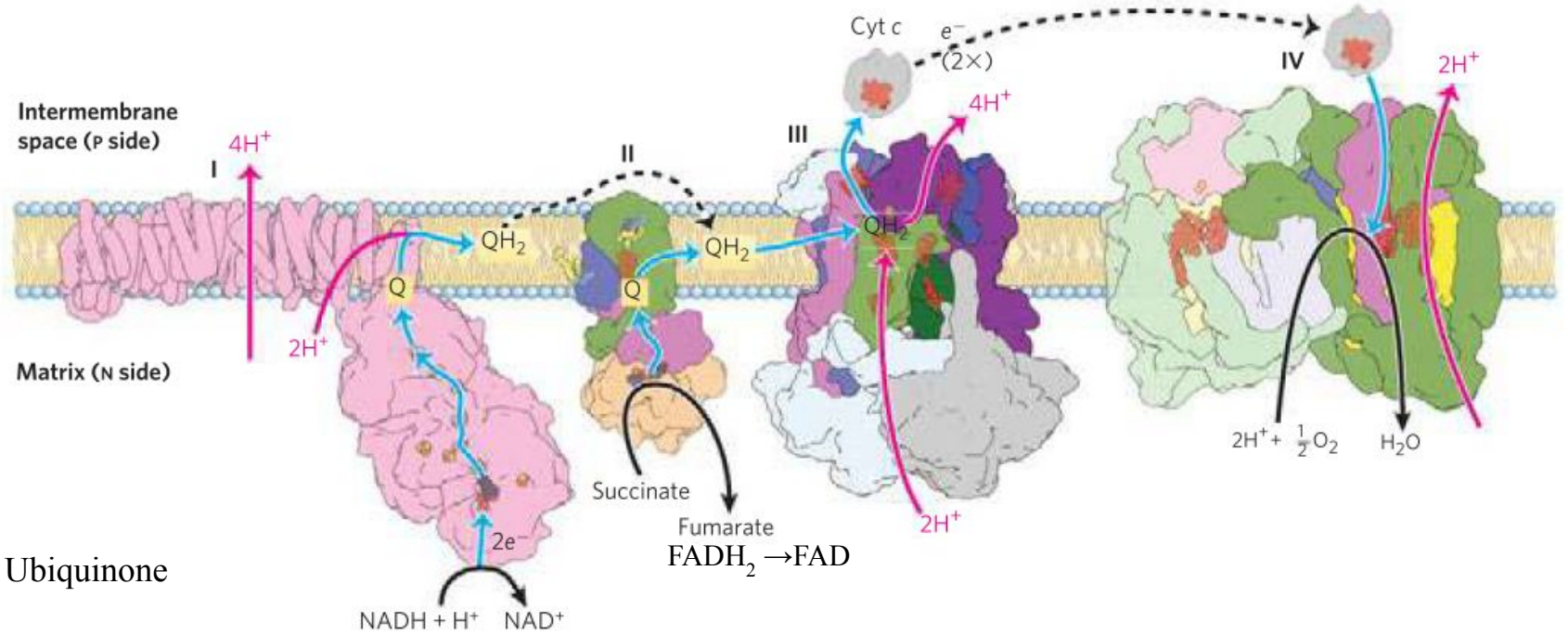
Step 2 of glucose metabolism: TCA Cycle (The Citric Acid Cycle)





TCA cycle takes place in Mitochondria.

Step 3 of glucose metabolism: Oxidative Phosphorylation (in mitochondria)
Flow of electrons and protons through the four complexes of the respiratory chain



Proton motive force

Q: Ubiquinone

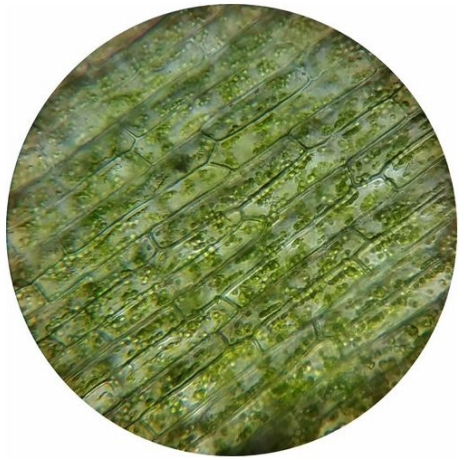
TABLE 19–6 Respiratory Proteins Encoded by Mitochondrial Genes in Humans

Complex	Number of subunits	Number of subunits encoded by mitochondrial DNA
I NADH dehydrogenase	43	7
II Succinate dehydrogenase	4	0
III Ubiquinone:cytochrome *c* oxidoreductase	11	1
IV Cytochrome oxidase	13	3
V ATP synthase	8	2

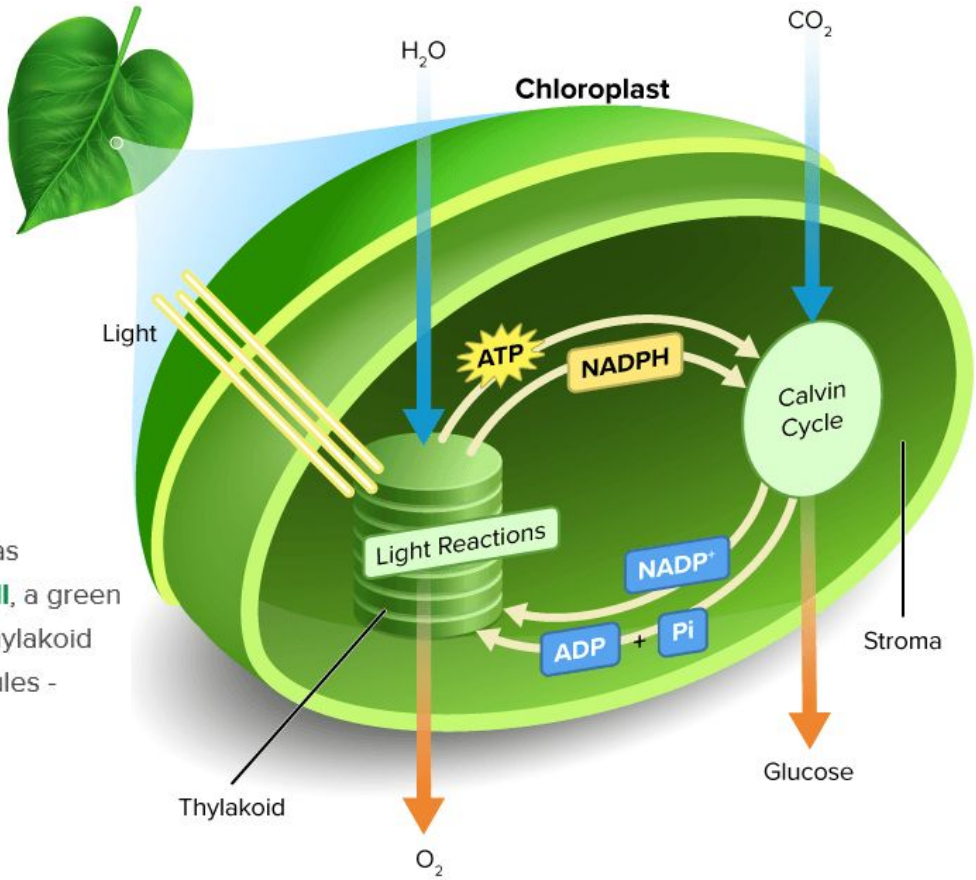
Chloroplast (kitchen of the cell)

- Double membrane-bound organelle. Present only in plant cell. Autonomous self-duplicating body. They have their own DNA and ribosomes.
- The endosymbiotic theory states that chloroplasts were once independently living bacteria.
- Chloroplast – Perform Photosynthesis – Helps in the release of oxygen.
- Photosynthesis occurs in two steps:
  1) Light Reaction: converts water and light into chemical energy stored in ATP and NADPH and releases Oxygen.
  2) Calvin cycle: combines carbon from CO_2 with chemical energy from ATP and NADPH to produce glucose.

Chloroplasts under the microscope in plant cells



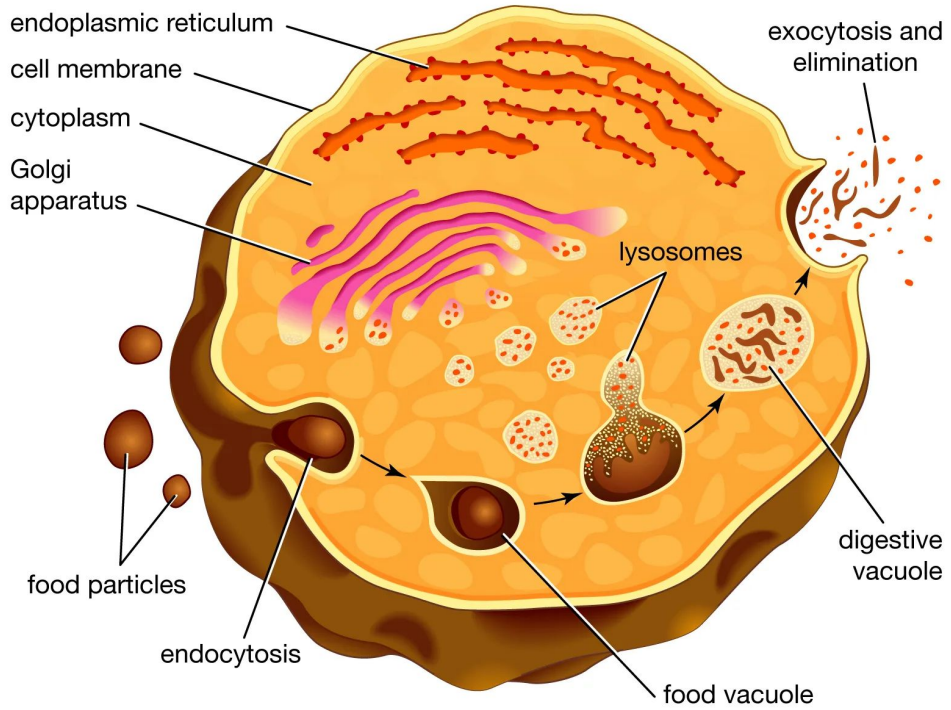
Each chloroplast contains neat stacks called **grana** (singular, granum). The grana consist of sac-like membranes, known as **thylakoid** membranes. These membranes contain **photosystems**, which are groups of molecules that include **chlorophyll**, a green pigment. The light reactions of photosynthesis occur in the thylakoid membranes. The **stroma** is the space outside the thylakoid membranes. This is where the reactions of the Calvin cycle take place. In addition to enzymes, two basic types of molecules - pigments and **electron carriers** – are key players in this process and are also found in the thylakoid membranes.



https://youtu.be/YxUG5EuBI-E

Lysosomes (Suicidal bags of the cell)

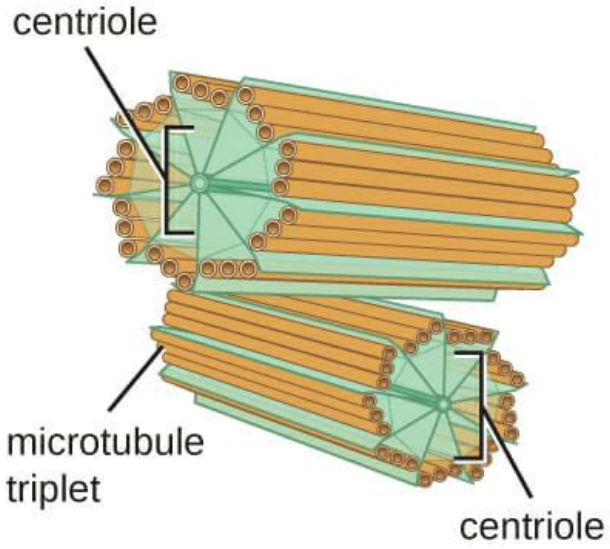
- Lysosomes are membranous sacs that **hydrolyze macromolecules** to carry out intracellular digestion. This may occur for a variety of reasons. Single-celled organisms, such as amoebas, use lysosomes to digest food products. This process is referred to as **phagocytosis**. Phagocytosis occurs in human cells as well, however in humans this process is used in defense to destroy invaders and bacteria.
- Lysosomes are also used to recycle the cell's own materials. This processes is referred to as **autophagy**. Damaged organelles that are broken down in the lysosome and its organic monomers are returned to the cell cytosol for reuse. In this way the cell is constantly renewing itself.



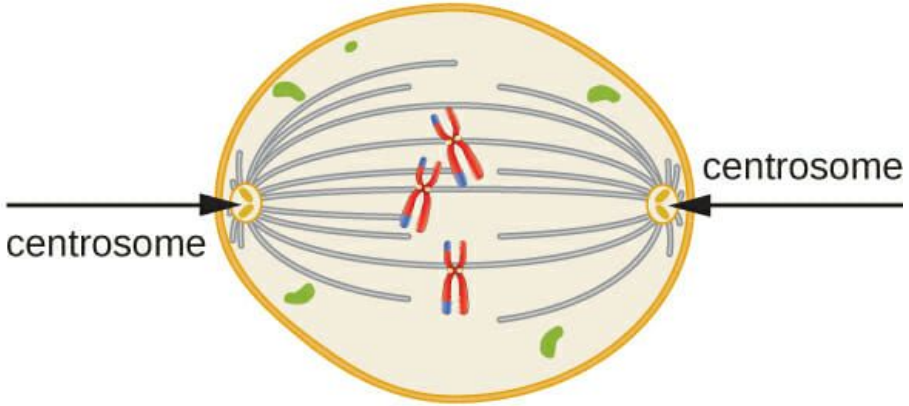
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Centrioles (Centrosomes)

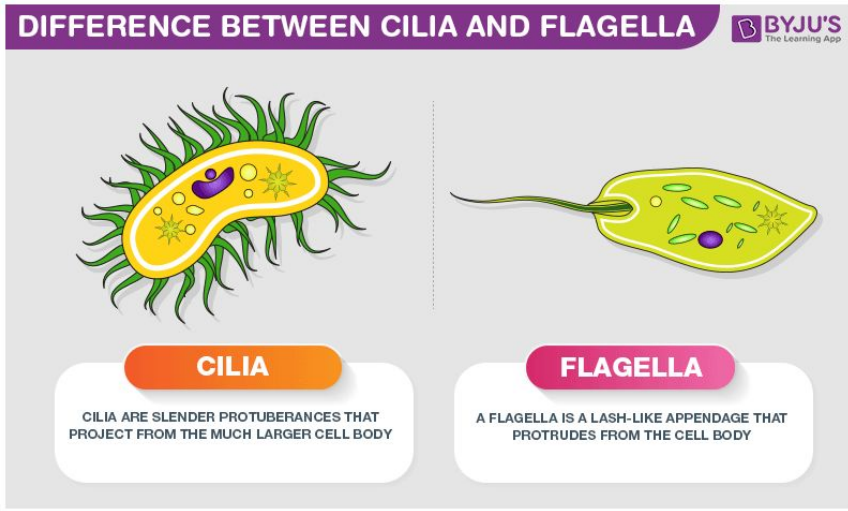
- Centrioles are non-membranous organelles.
- Centrioles are structural organelles consisting of nine microtubule triplets organized into cylinders.
- The two main functions of centrioles are the formation of basal bodies and mitotic spindles.
- **Basal bodies** are used as building blocks for flagella and cilia.
- **Mitotic spindles** are involved in the separation of chromosomes during cell division. Centrioles determine the location of mitotic spindles during anaphase.



(a)



(b)



Microtubules

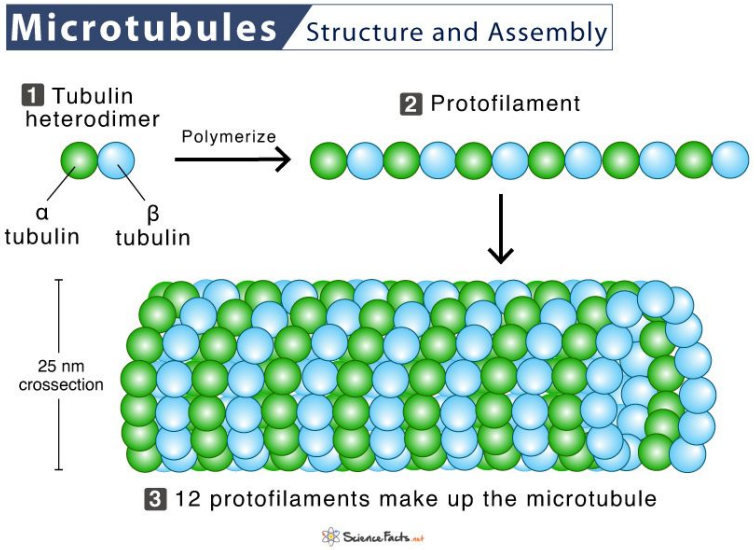
- Microtubules are non-membranous organelles. They are involved in the **movement of organelles** and other structures, for example lysosomes, mitochondria and chromosomes.

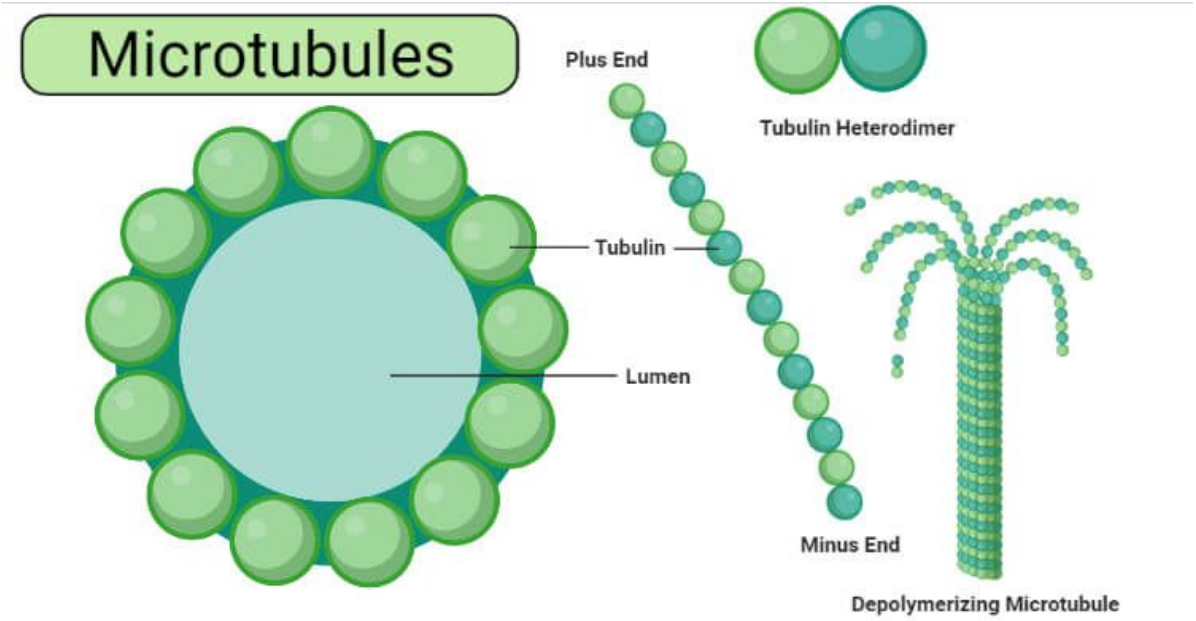
Watch this video to see how motor proteins (kinesin) walk on the microtubule. https://www.youtube.com/watch?v=y-uuk4Pr2i8

- They are elongated, nonbranching polymers made up of dimers of **α-tubulin** and **β-tubulin**.

- Microtubules contain approximately 13 circular dimeric tubulin molecules. Dimers can be added or removed to change the length of the microtubule. This process is termed **dynamic instability** and requires **GTP hydrolysis**.

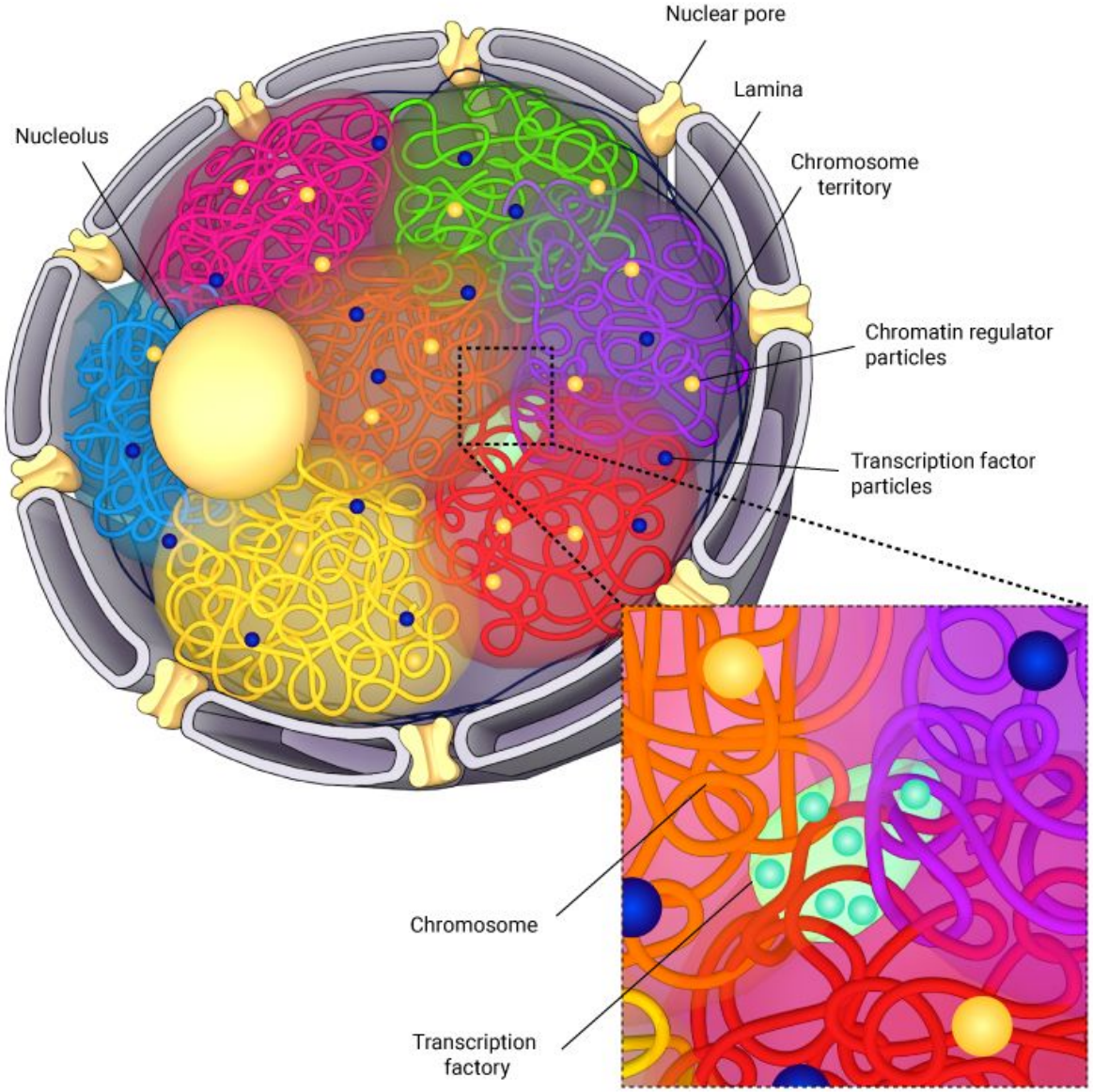
- All of the tubulin dimers are arranged in a specific pattern so that they have the same orientation. Because of this orientation microtubules are considered polar, with a plus and minus end. Growth occurs at the plus end. The minus end of the microtubule does not grow.



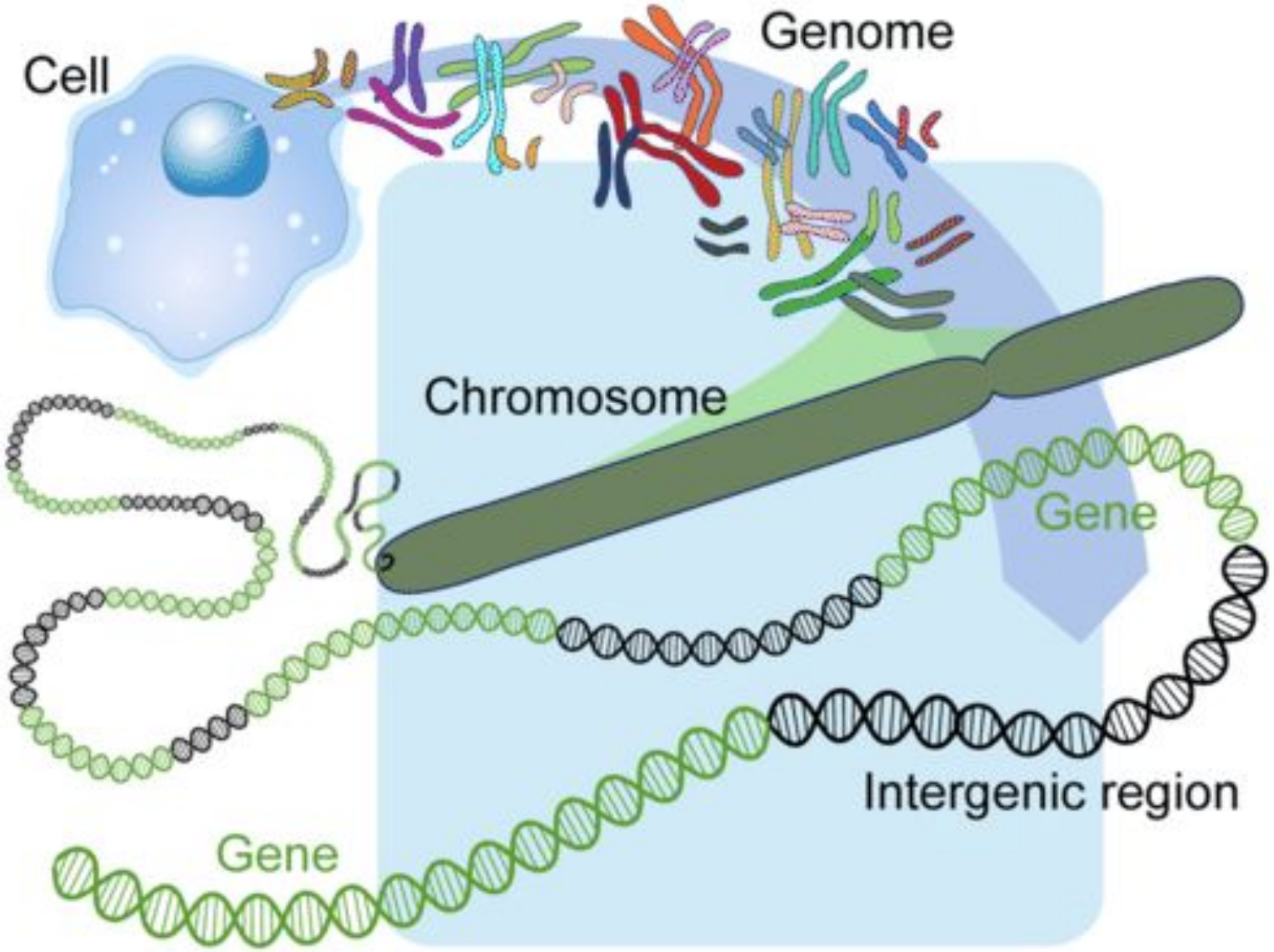


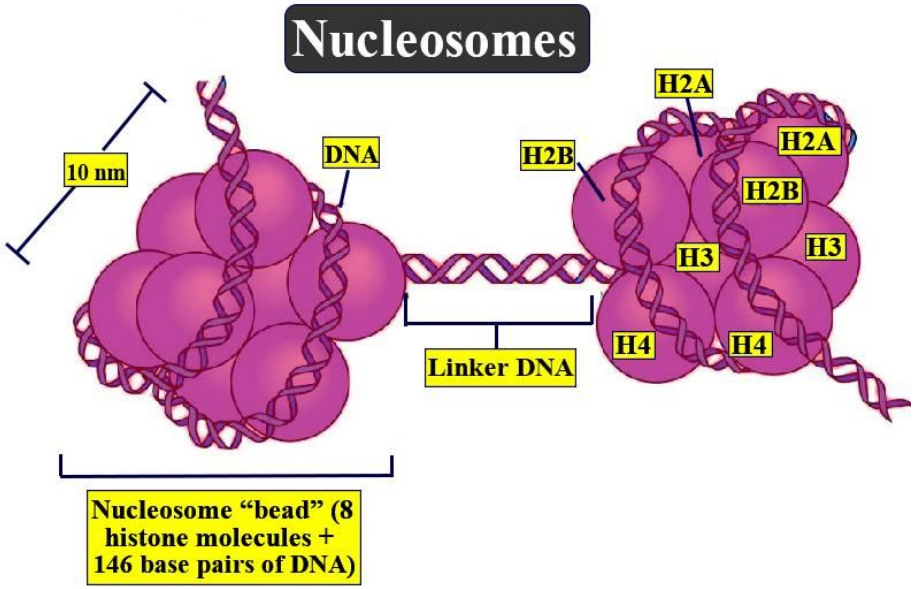
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Genome Organization and Nuclear Architecture



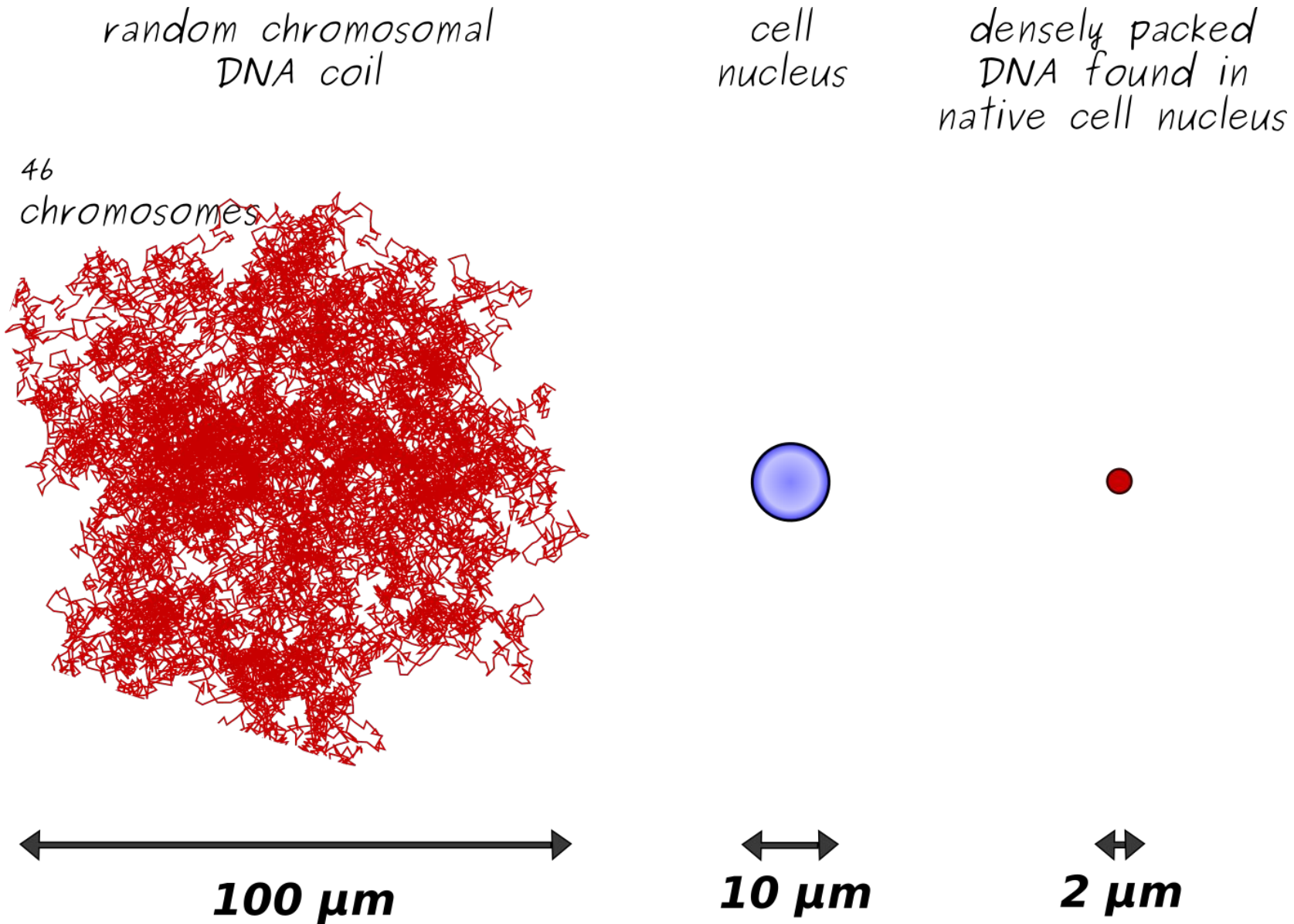
What is the difference between a Gene, a Chromosome and a Genome?



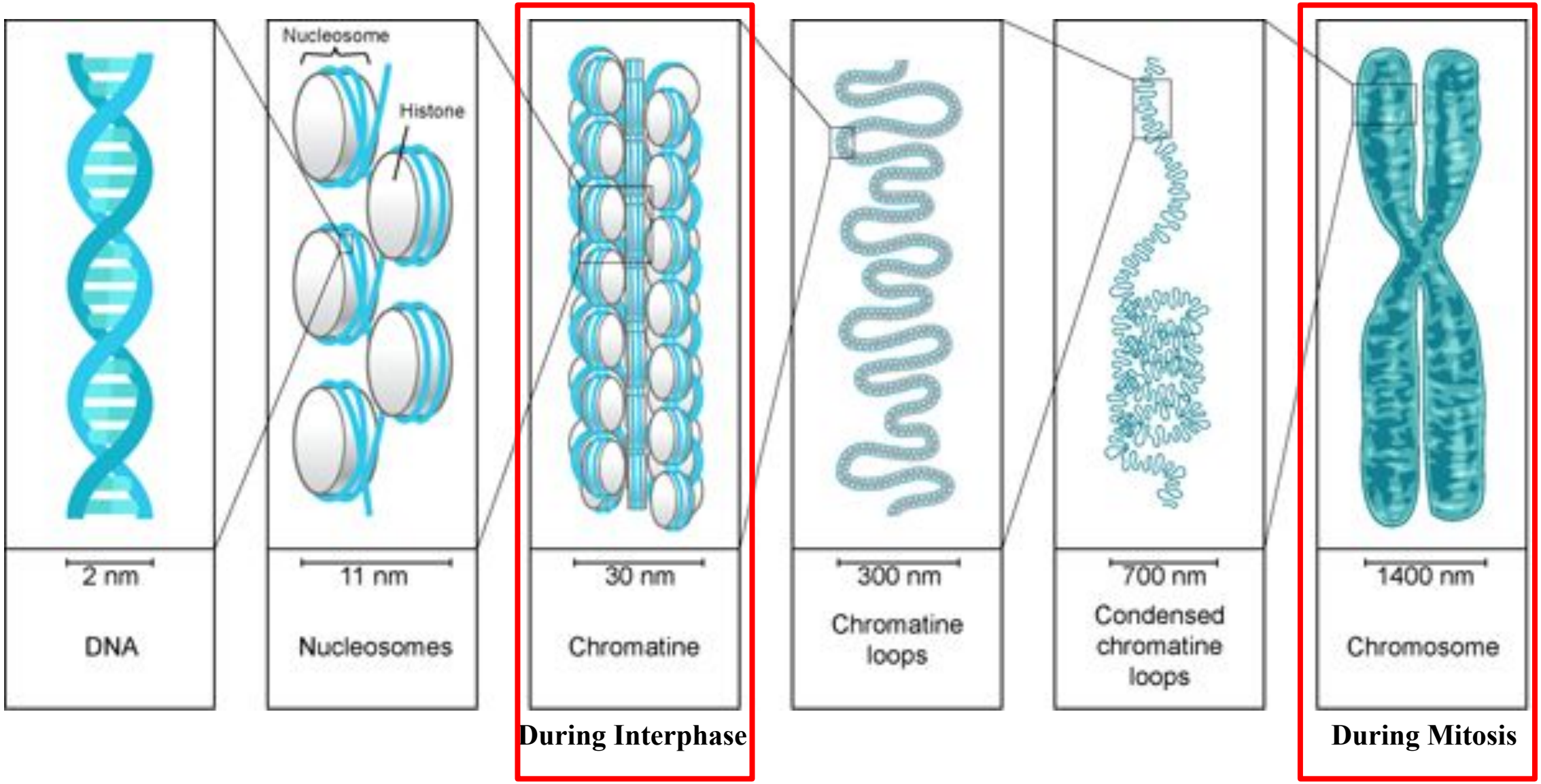


DNA in the interphase nucleus

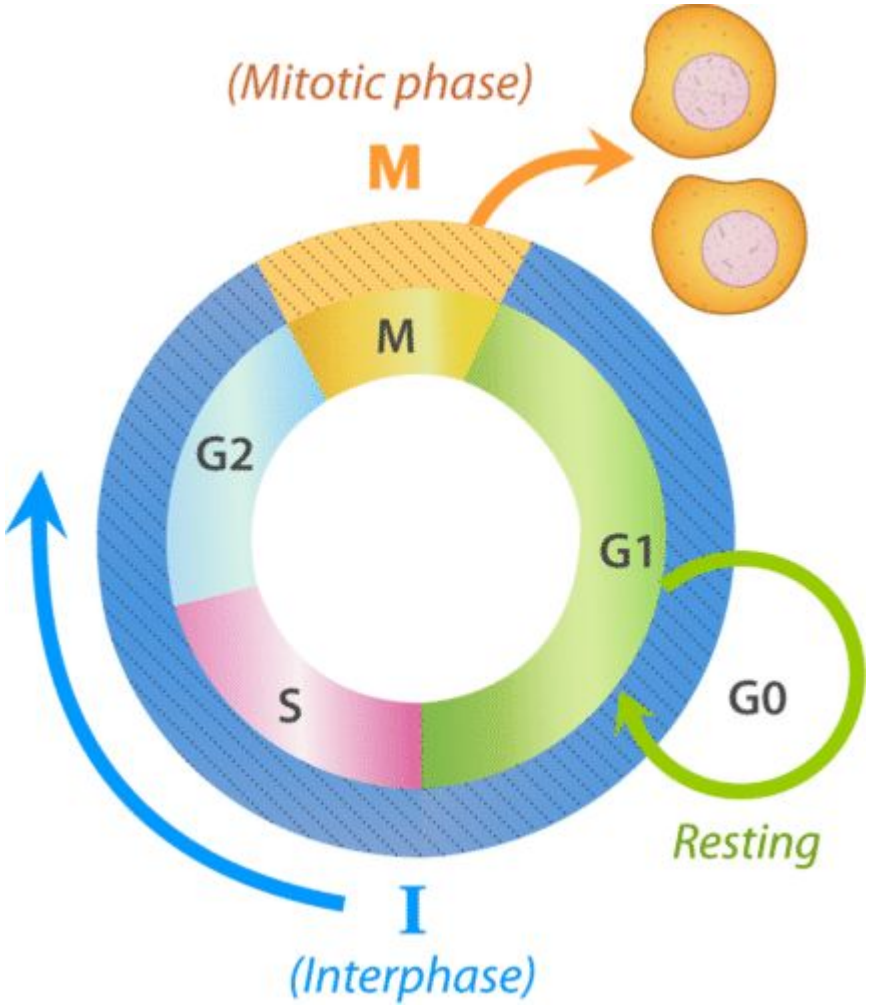
Human genome is 2 meter long DNA (all 46 chromosomes)

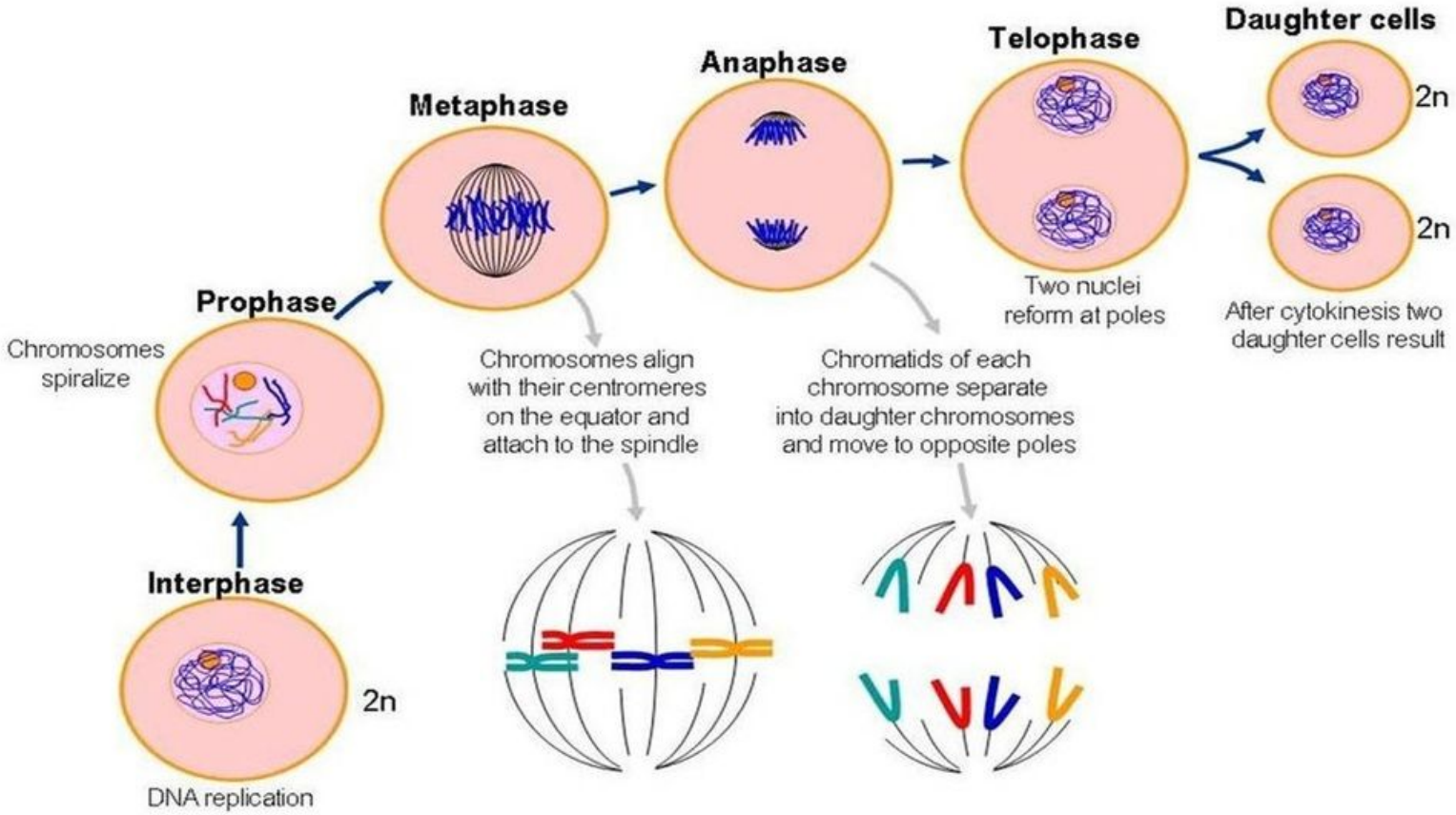


How DNA is packaged into chromosomes during mitosis (cell division)?

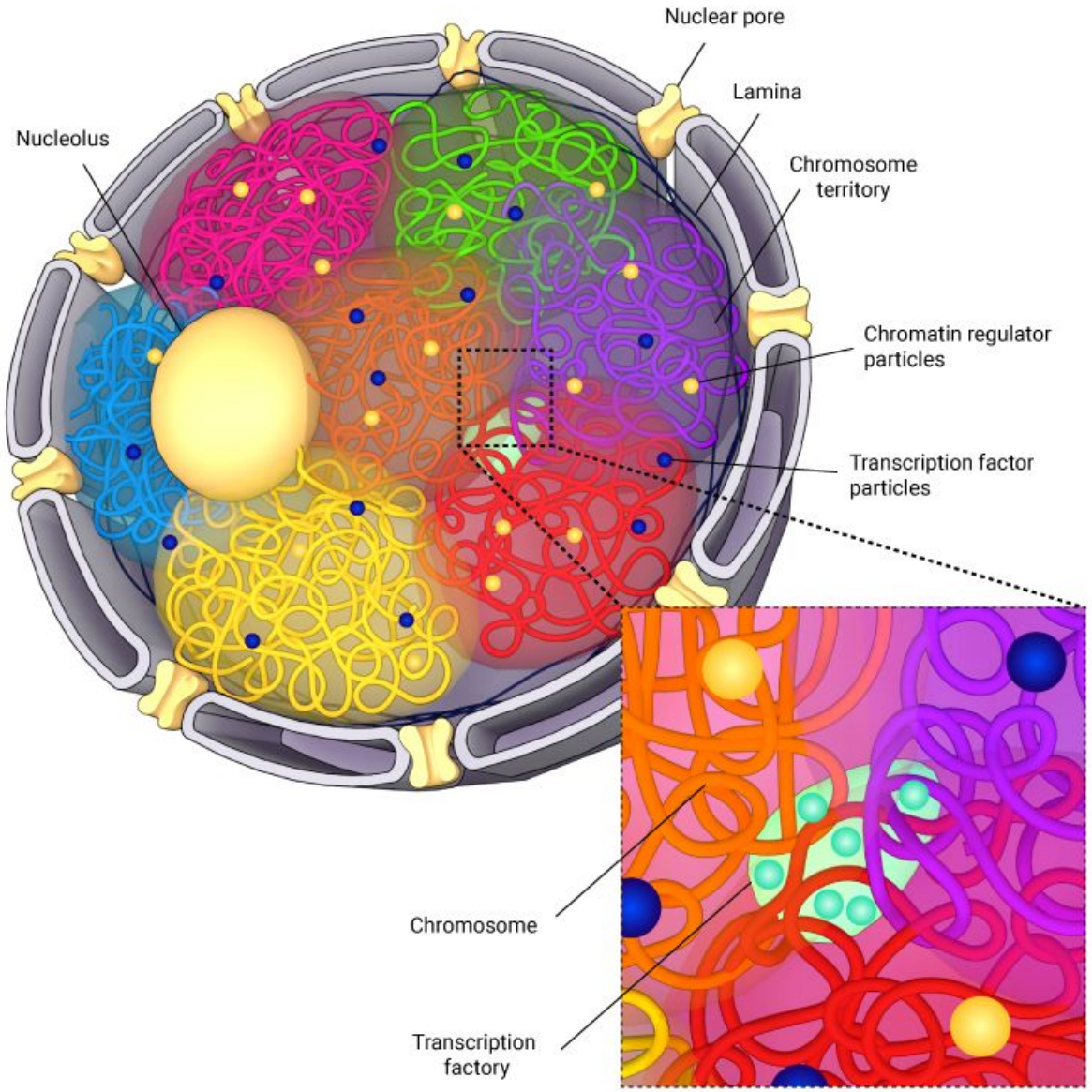


Cell Cycle: Interphase and Mitosis



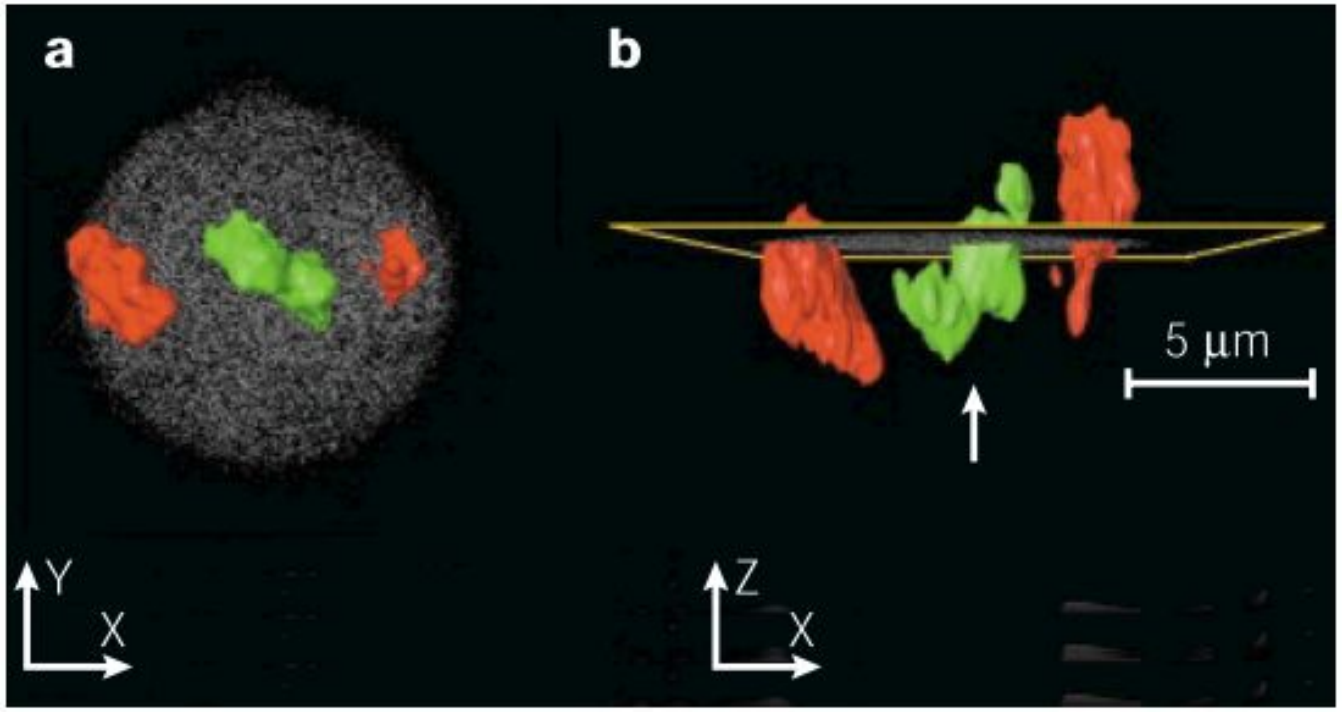


Territorial organization of chromosomes during interphase



How the chromosome territories are organized?
Is it a random organization or has some preference?

Chromosome 18 Gene Poor CTs: Red
Chromosome 19 Gene Rich CTs: Green



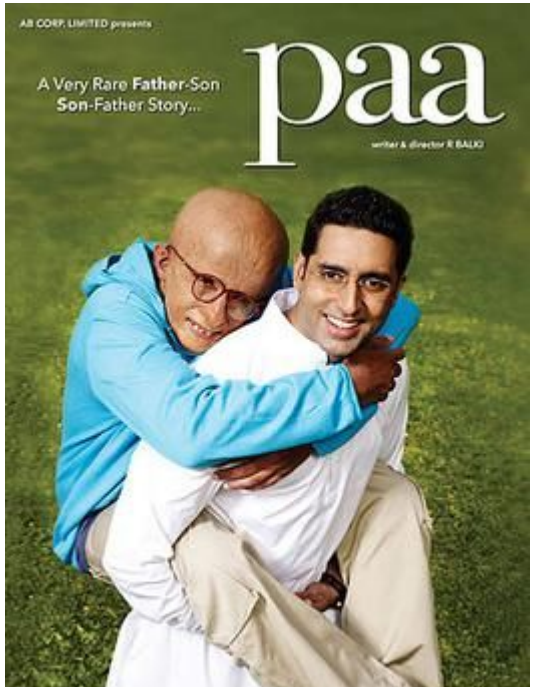
Both chromosomes have same DNA content, still different preference for nuclear localization.

Figure 4 | **Gene-rich and gene-poor chromosome territories.** Three-dimensional reconstructions of chromosome 18 (red; gene-poor) and 19 (green; gene-rich) territories painted in the nucleus of a non-stimulated human lymphocyte. (Image courtesy of Marion Cremer and Irina Solovei.) Chromosome 18 territories were typically found at the nuclear periphery, whereas chromosome 19 territories were located in the nuclear interior[42]. **a** | X,Y view: a mid-plane section of the nucleus is shown as a grey shade. Only the parts of the territories below this section can be seen. **b** | X,Z view: the arrow marks the side from which the section in part **a** is viewed.

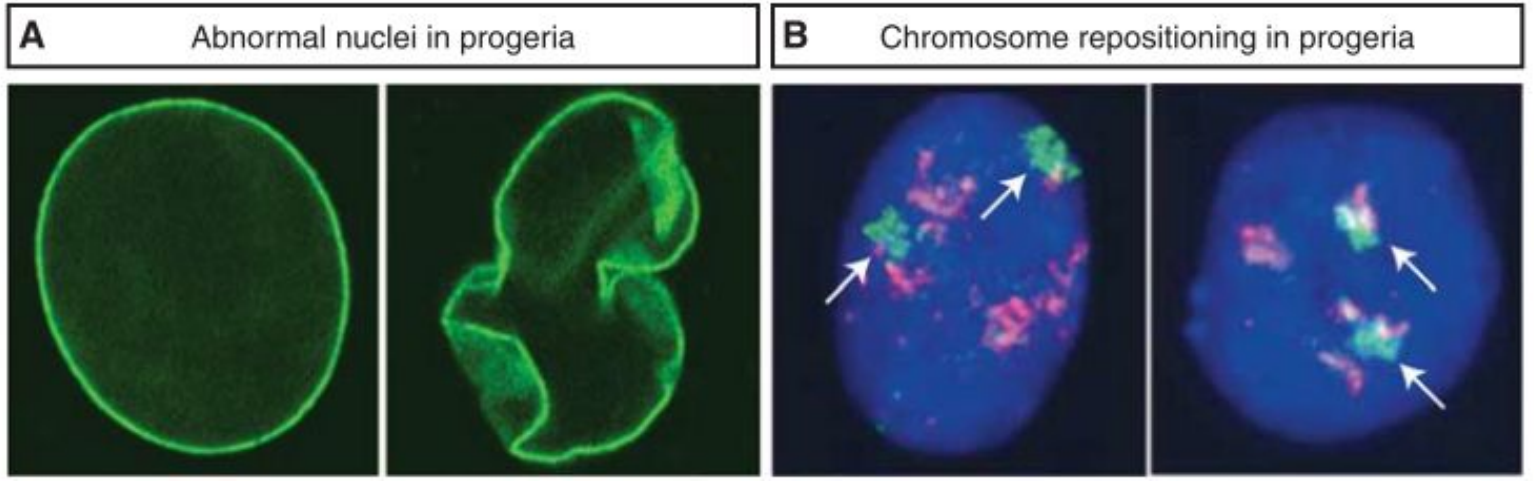
Croft et al. 1999

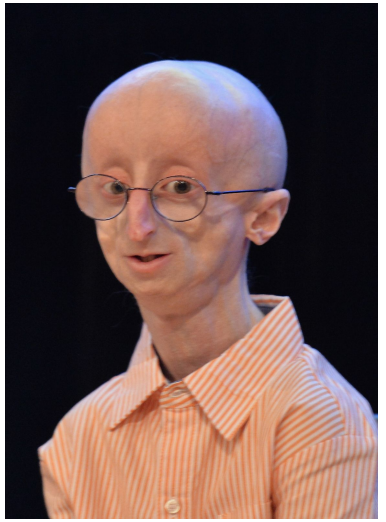
Altered gene positioning in diseases

- Lamins and emerin (EMD) are integral proteins of the nuclear envelope and are linked to a group of rare genetic disorders called nuclear envelopathies. (Webster et al. 2009)
- Well known nuclear envelopathies such as Hutchinson-Gilford progeria syndrome (HGPS, premature aging) and Emery-Dreifuss muscular dystrophy (EDMD) are associated with mutations of the lamin A/C or emerin genes and are often characterized by alterations in nuclear morphology and may compromise gene expression.
- Studies in primary fibroblasts of several nuclear envelopathies (including HGPS and EDMD) show repositioning of chromosomes 13 and 18 toward the nuclear interior. (Meaburn et al. 2007)









Altered gene positioning in diseases

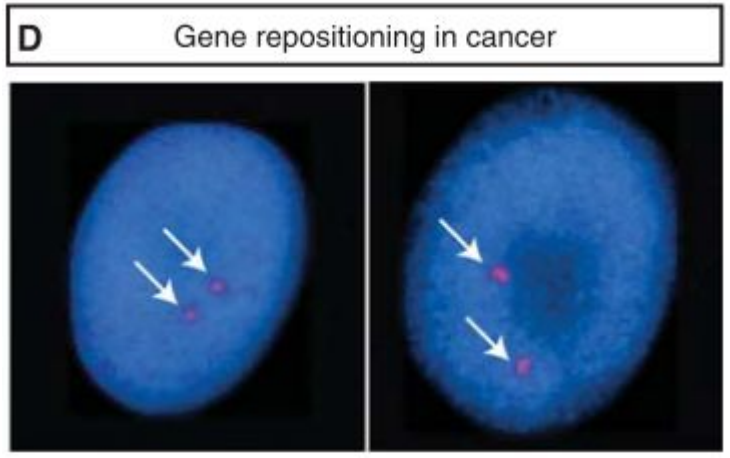
- Altered chromosome positioning has also been associated with numerous types of cancer in which altered gene expression patterns lead to malignant transformation of the cell.

- Chronic leukemias are associated with enlarged nuclei in lymphoid cells.

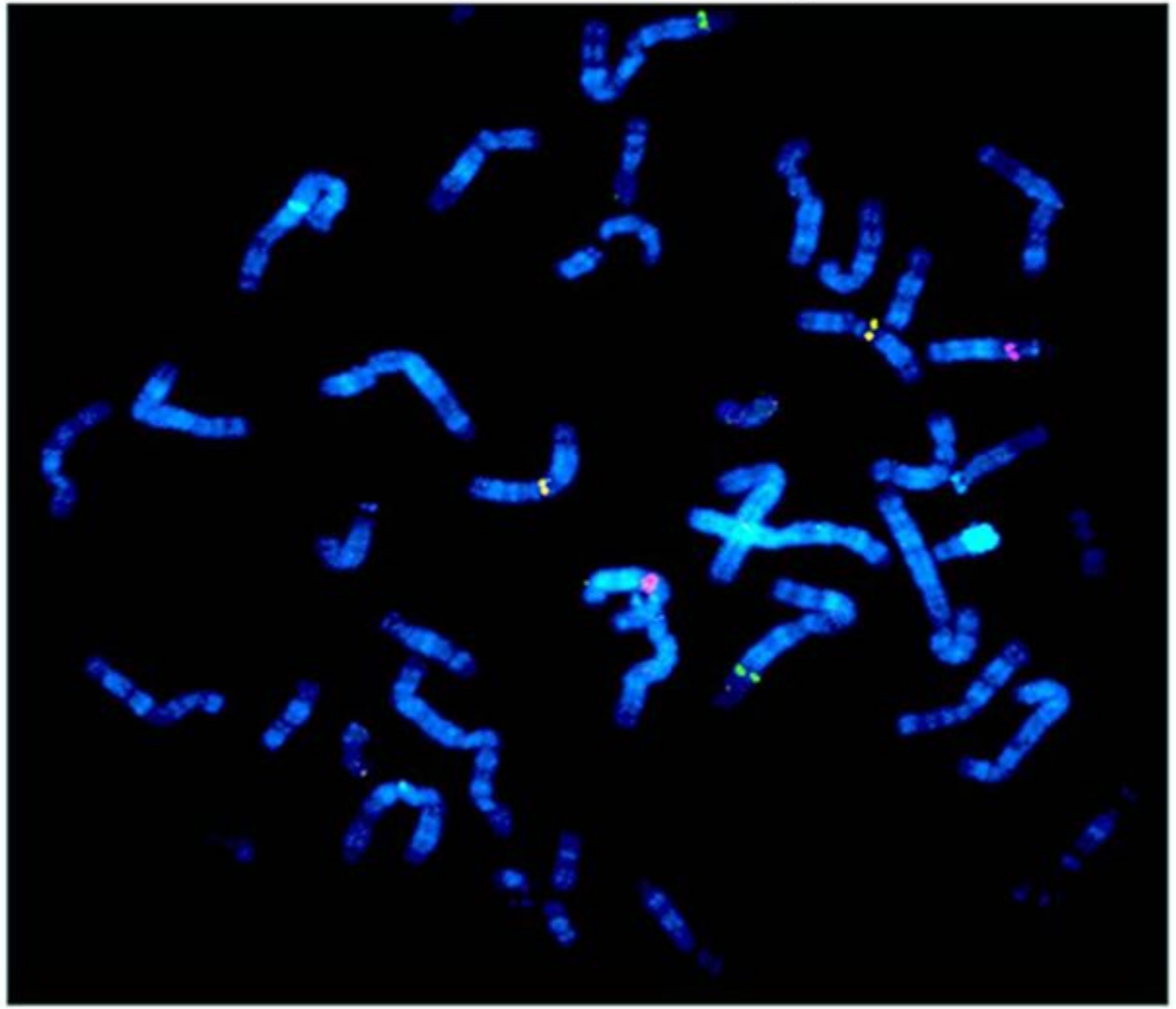
- Abnormal relocation of chromosome 18 from the nuclear periphery to the interior has been documented in several types of tumor cell lines, including melanoma-derived line, cervix carcinoma, colon carcinoma, Hodgkin-derived cell line and colon carcinoma metastatic cells.

- A radial shift of chromosome 8 and a significant change in its shape was observed in pancreatic neoplastic tissue as well as a radial shift of centromeric regions of chromosome 17 and HER2 domains in tumor breast tissues.

- In vitro induction of tumorigenesis in early breast cancer showed altered positioning of a set of cancer-associated genes such as AKT1, BCL2, ERB2, VEGF, although no correlation was found between radial redistribution and gene activity level.

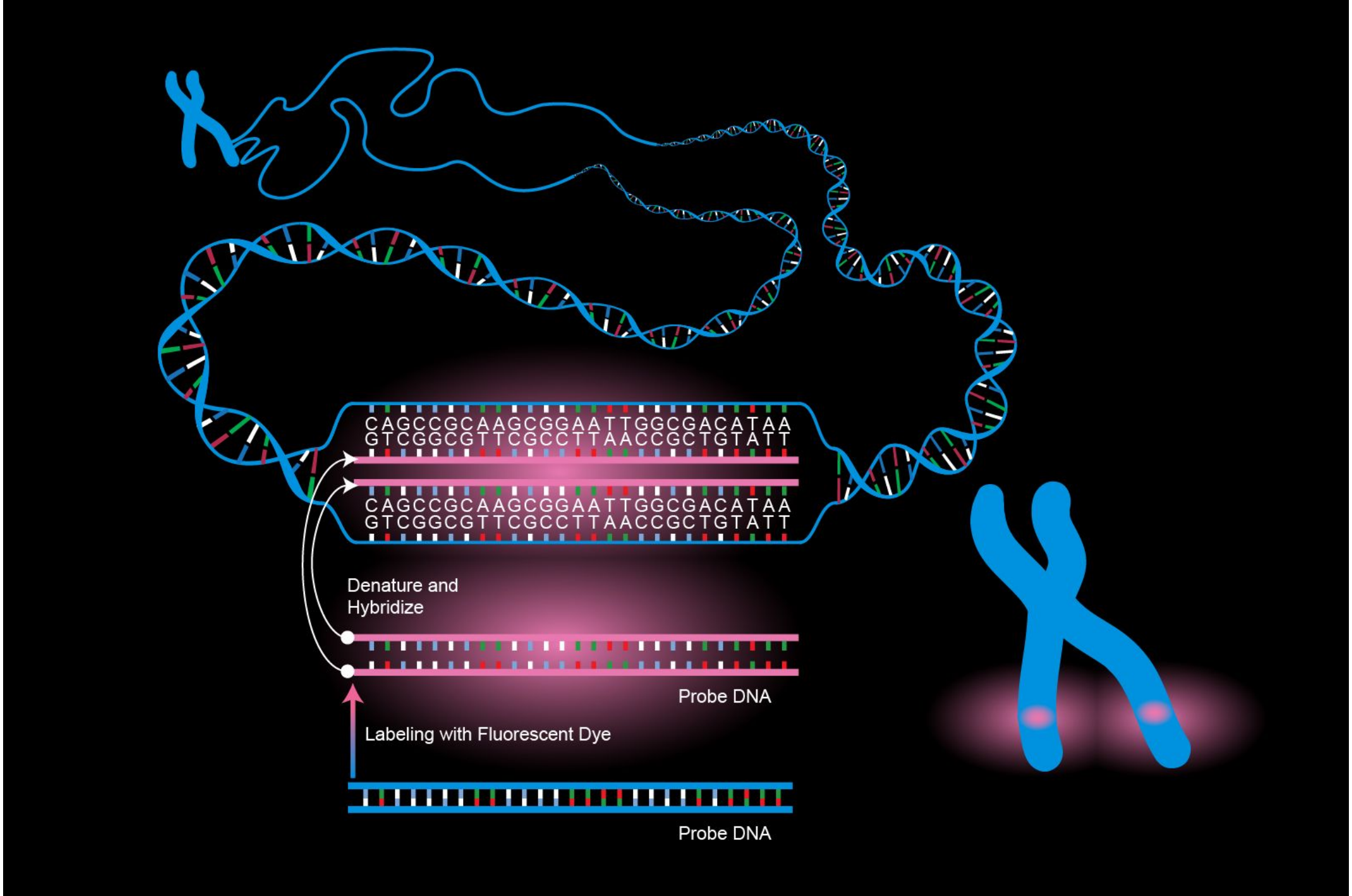


Fluorescence *In-situ* Hybridization (FISH)

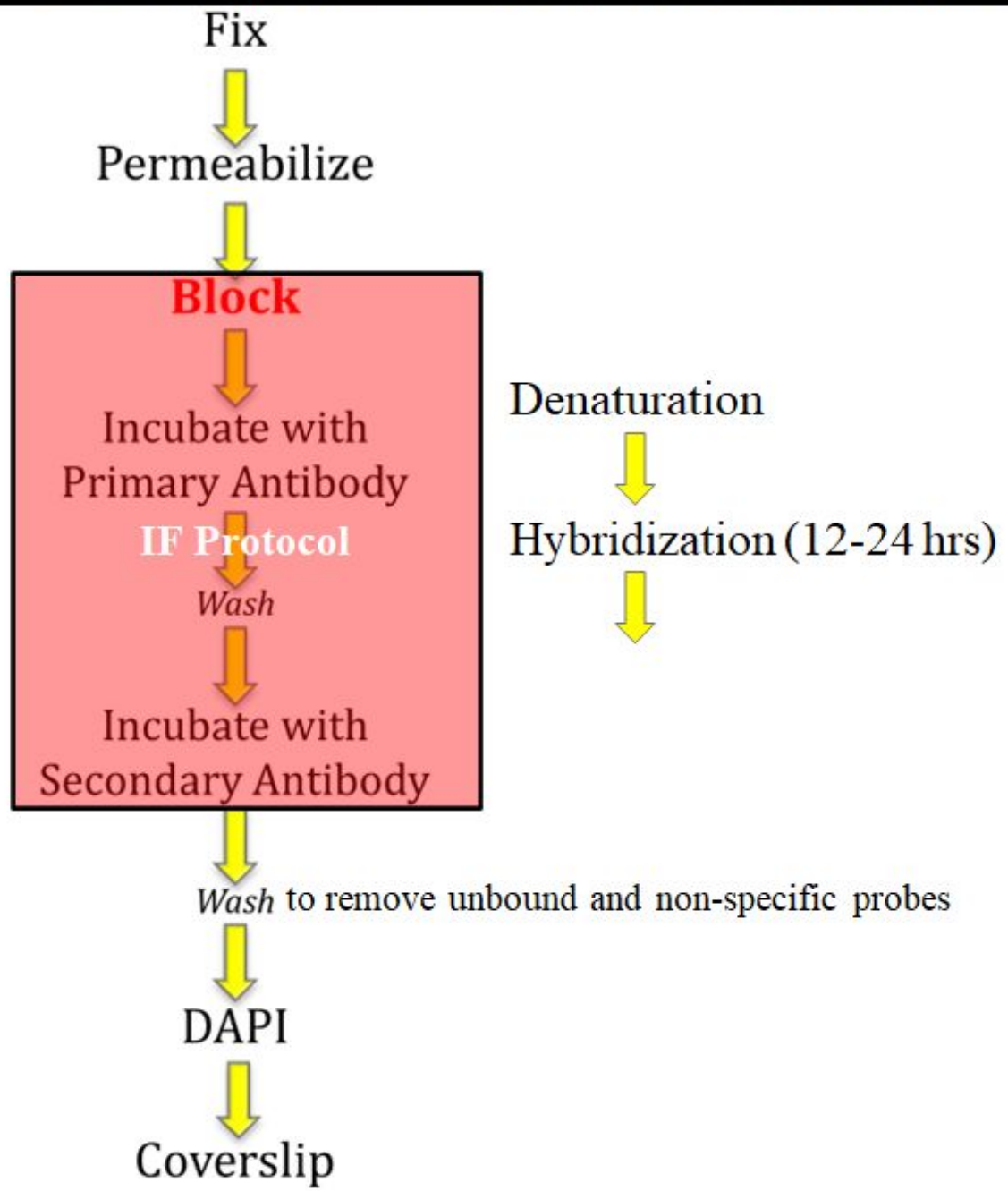


Fluorescence in situ hybridization (FISH) is a laboratory technique for detecting and locating a specific DNA sequence on a chromosome.

The technique relies on exposing chromosomes to a small DNA sequence called a probe that has a fluorescent molecule attached to it. The probe sequence binds to its corresponding sequence on the chromosome.


CAGCCGCAAGCGGAATTGGCGACATAA
GTCGGCGTTCGCCTTAACCGCTGTATT
CAGCCGCAAGCGGAATTGGCGACATAA
GTCGGCGTTCGCCTTAACCGCTGTATT
Denature and
Hybridize
Probe DNA
Labeling with Fluorescent Dye
Probe DNA

In-Situ Hybridization (ISH) or Fluorescence in Situ Hybridization (FISH)



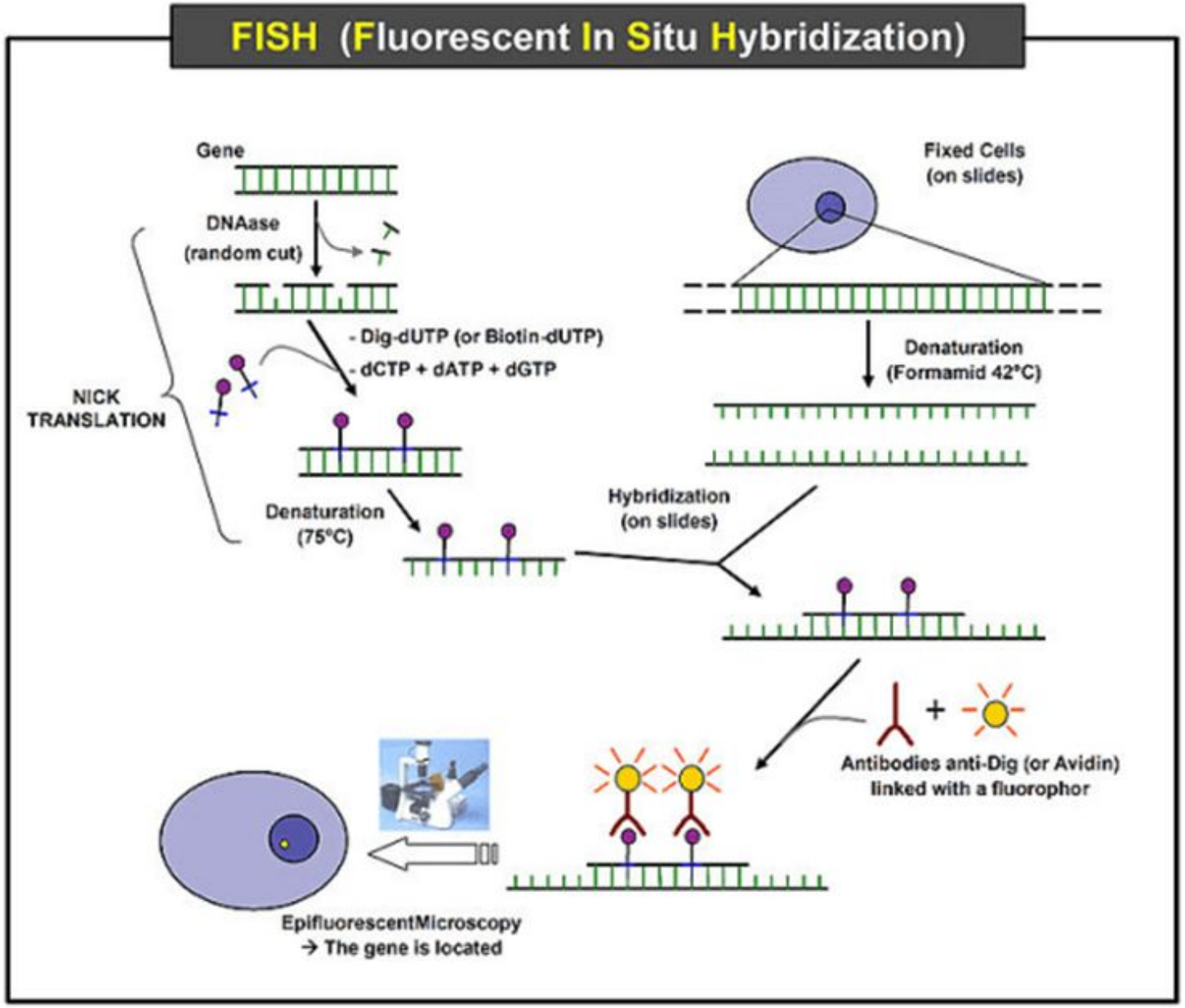
ISH or FISH is a technique that is used for localization and detection of **specific DNA** and **RNA** sequences in fixed cells and tissues by hybridizing the complementary strand of a nucleotide probe to a particular sequence.

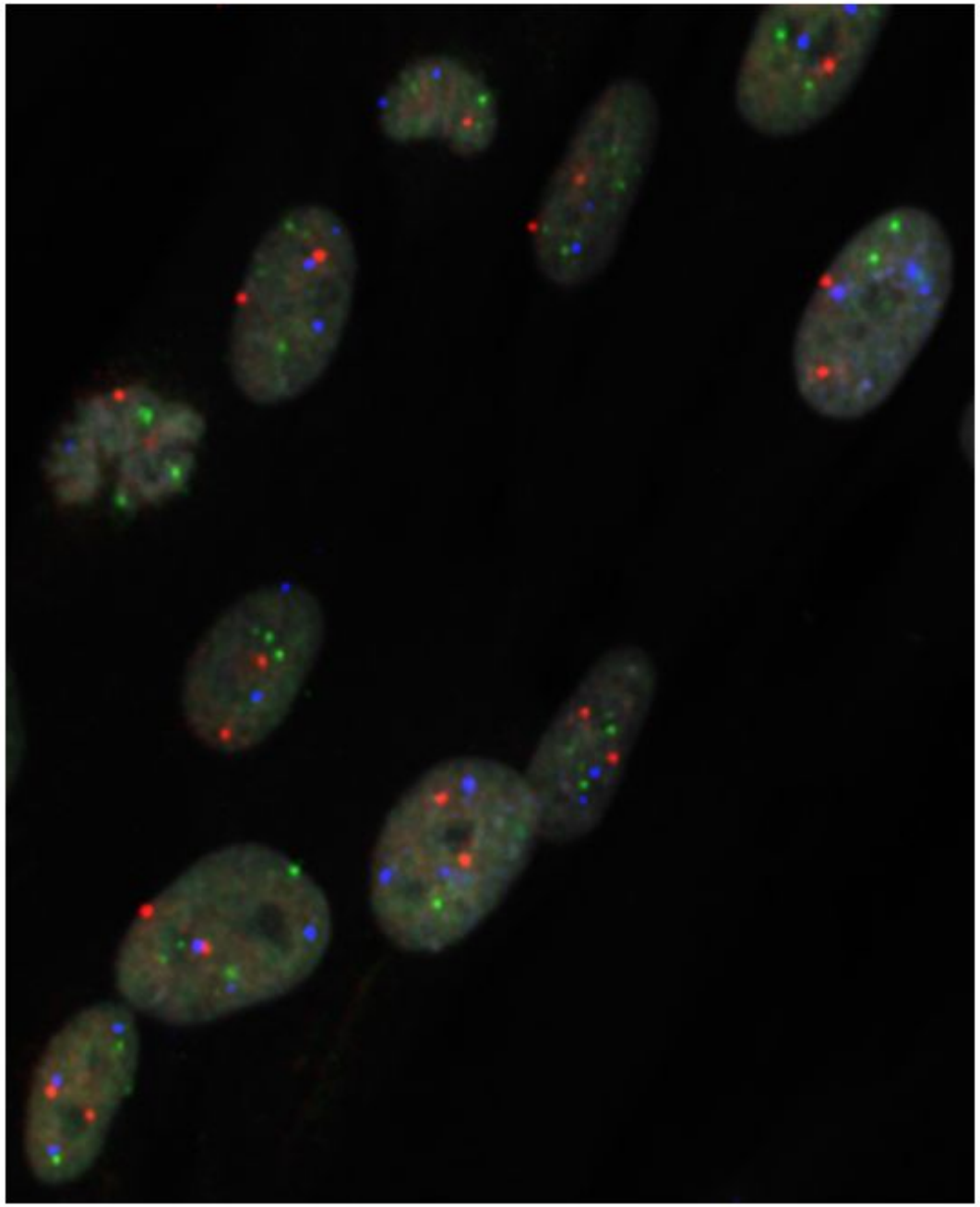
Fluorescent probes (~20 bp DNA/RNA conjugated with fluorescent dye) bind to the target DNA/RNA by complementary base pairing.

Applications:

- Karyotyping: Chromosomal aberrations
- Gene positioning in the nucleus
- Gene expression profiling

Multiplexed DNA FISH





Further reading

Review Articles:

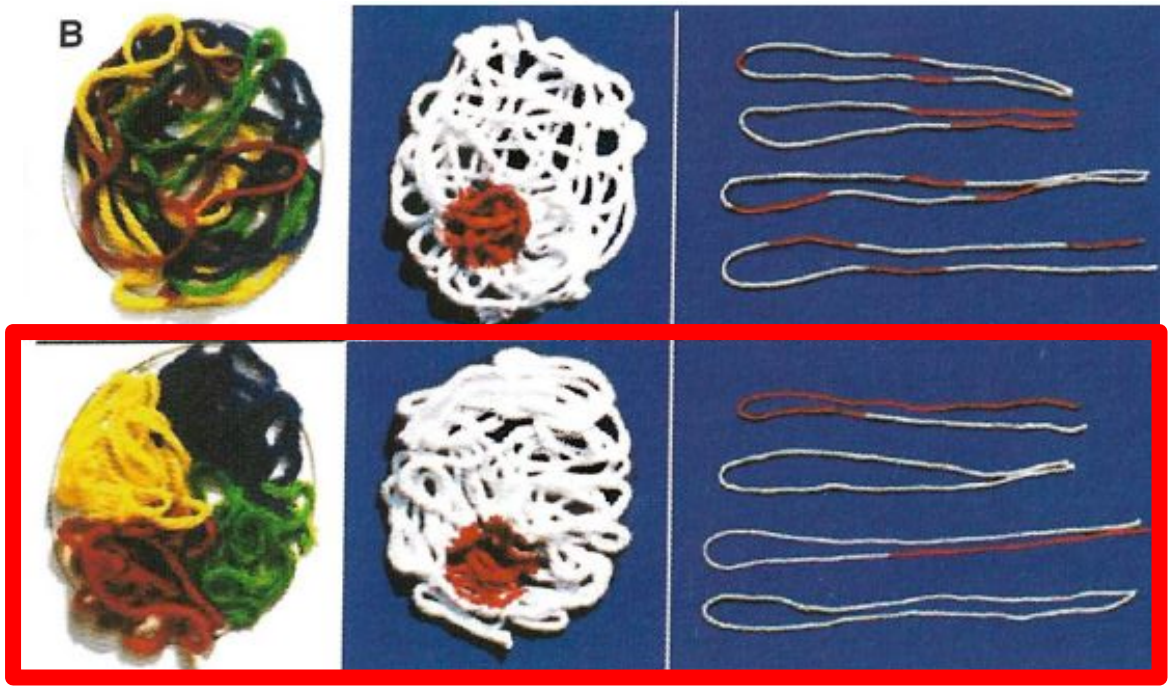
1) Branco MR & Pombo A. Intermingling of chromosome territories in interphase suggests role in translocation and transcription-dependent associations. PLoS Biology 2006; 4(5):e138. DOI: 10.1371/journal.pbio.0040138
2) Cremer T & Cremer C. Chromosome territories, nuclear architecture and gene regulation in mammalian cells. Nature 2001;2:292-301. DOI: 10.1038/35066075
3) Meaburn KJ & Misteli T. Cell Biology: Chromosome Territories. Nature 2007;445(7126):379-789. DOI: 10.1038/445379a
4) Cremer T & Cremer M. Chromosome Territories. Cold Spring Harbor Perspective in Biology 2010;2:a003889. DOI: 10.1101/cshperspect.a003889.

Next class on 05/11/2025 (Wednesday)

Laser-UV-microirradiation experiment to answer whether the chromosome organization is non-territorial or territorial.

Non-Territorial organization of chromosomes

Territorial organization of chromosomes



Steps of the Laser-UV-microirradiation experiment:

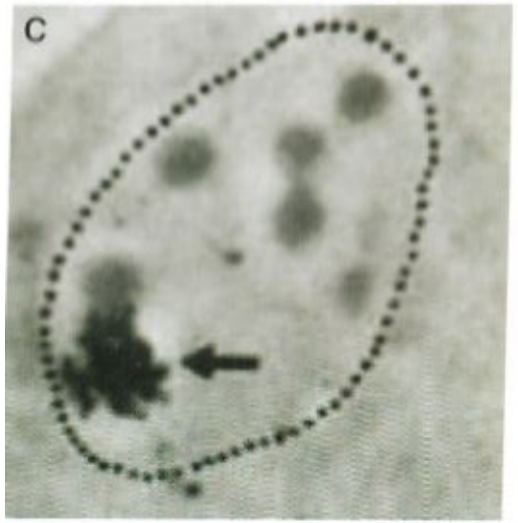
1) The nucleus of the live cell was microirradiated in G1

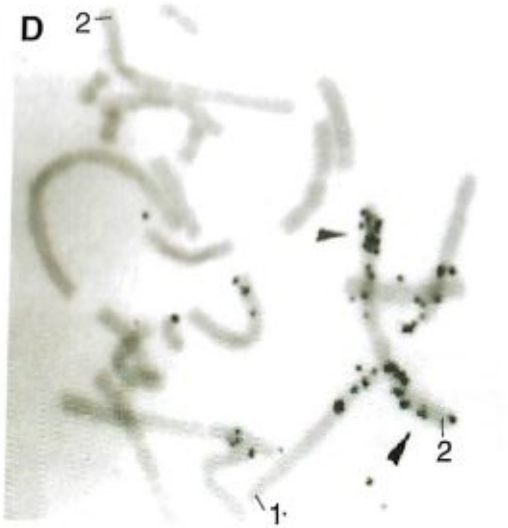
2) Pulse-labeled with 3H thymidine

3) Fixed immediately

4) Autoradiography

Autoradiograph of a diploid CHO cell nucleus after microirradiation and pulse-labelling

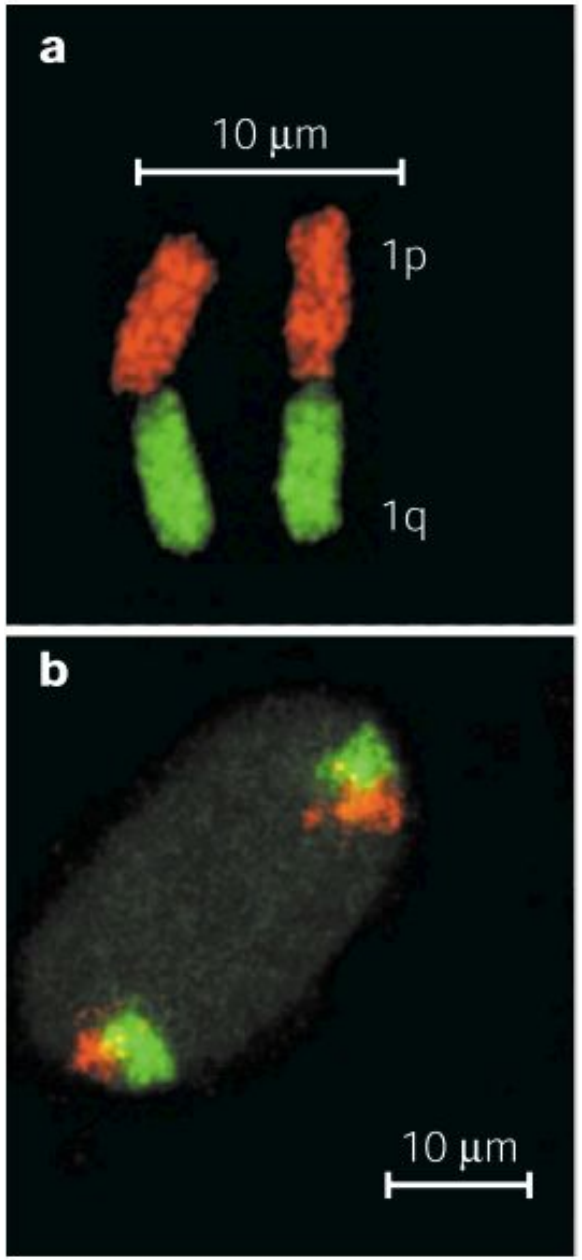




Metaphase spread from the same experiment

Zorn et al. 1976
Zorn et al. 1979
Cremer et al. 1982
Cremer et al. 1982

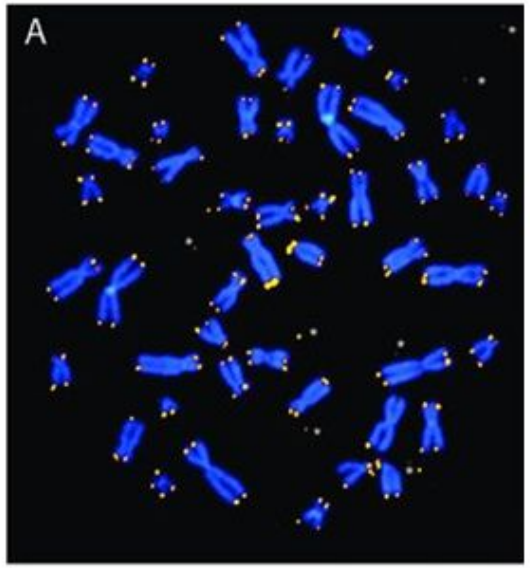
Features of human chromosome territories



Metaphase chromosome

Interphase chromosome territories

Types of probes for DNA FISH



Lansdorp 2022

PROBES

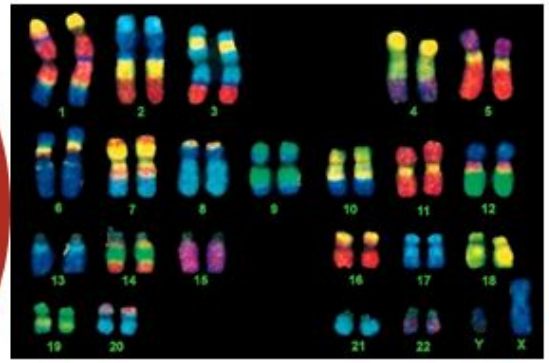
Centromere Probes
- Alpha and Satellit III Probes
- Generated from repetitive sequences found in centromeres
- Centromere regions are stained brighter



Hemmat et al. 2012

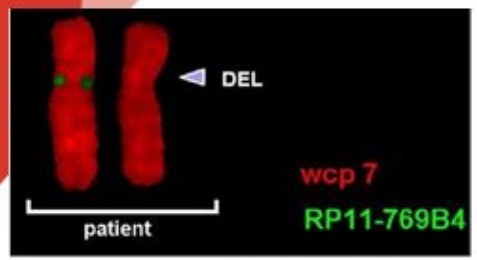
Telomere
- Specific for telomeres
- Specific to the 300kb locus at the end of specific chromosome

Whole Chromosome
- Collection of probes that bind to the whole length of chromosome
- Multiple probe labels are used
- Hybridize along the length of the chromosome



https://www.mun.ca/biology/scarr/FISH_chromosome_painting.html
© Steven M. Carr

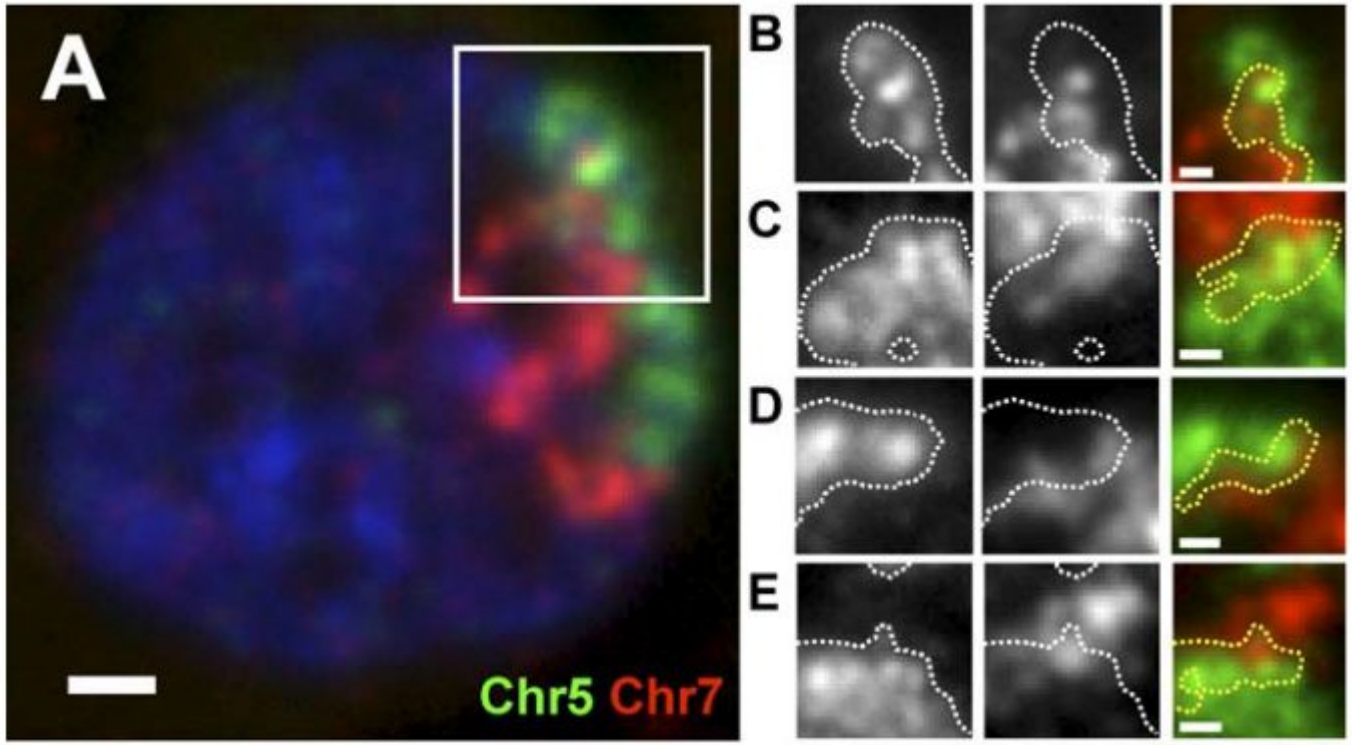
Locus
- Deletion
- Translocation probes
- Gene detection and localization probes
- Gene amplification probes

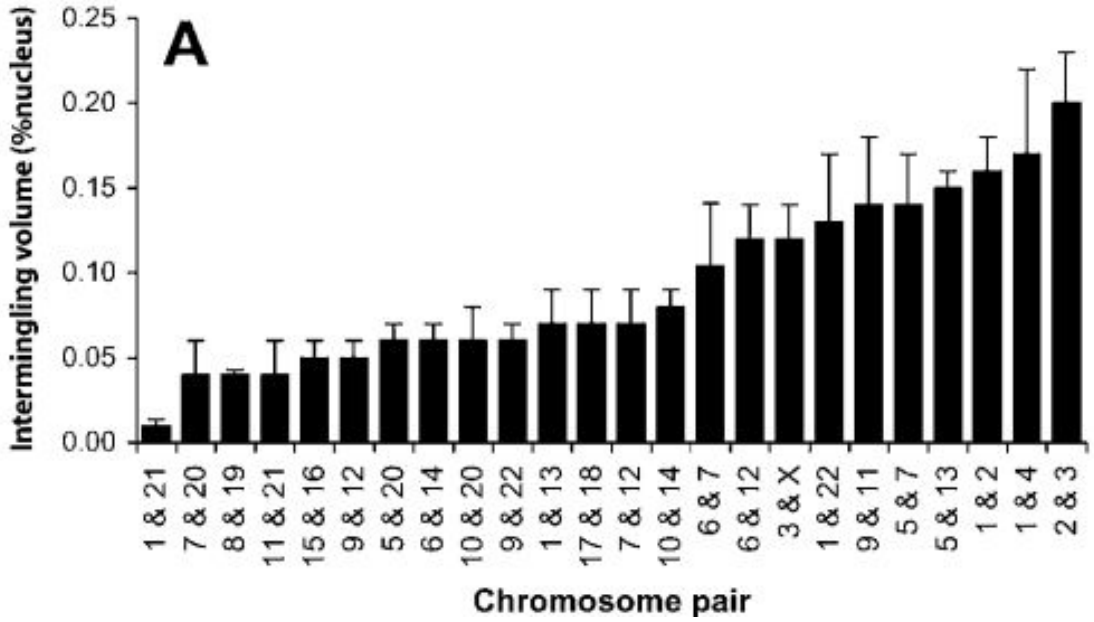


Varvagiannis et al. 2014

Intermingling of chromosome territories revealed by Cryo-FISH

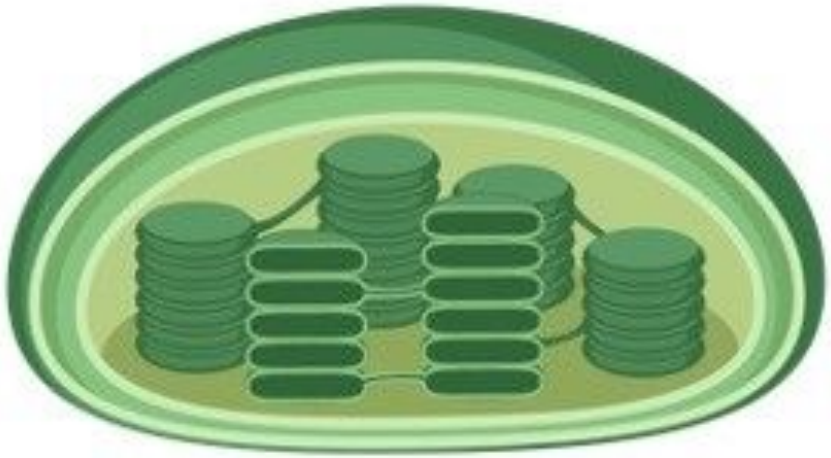
Intermingling of chromosome territories





Branco and Pombo 2006

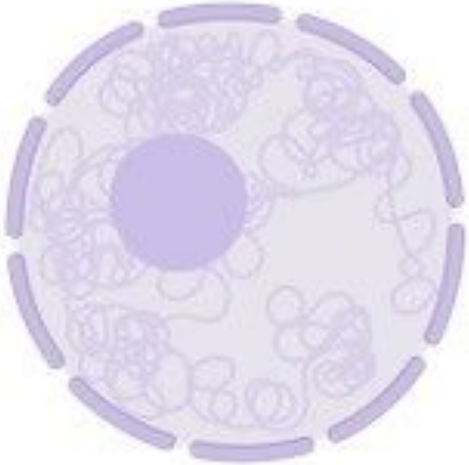
Chloroplast



Golgi Apparatus

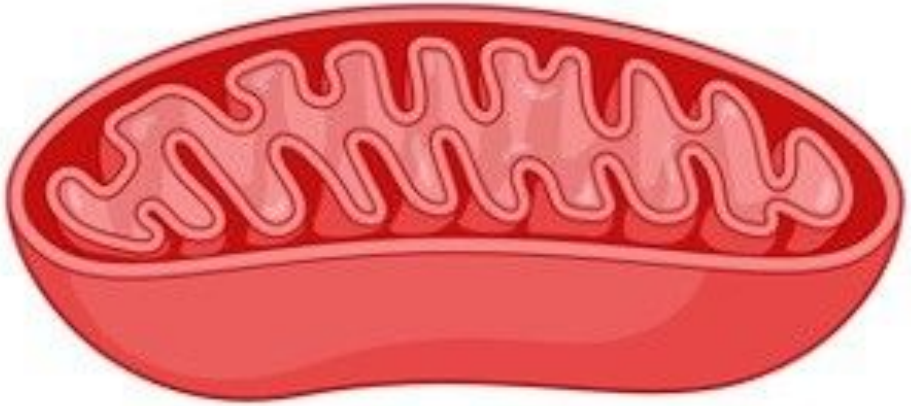


Nucleus

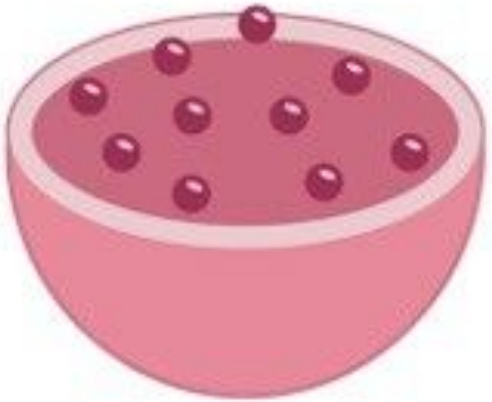


Cell Organelles

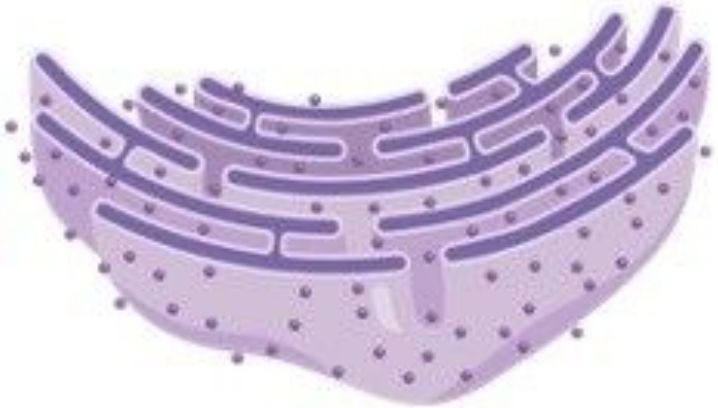
Mitochonria



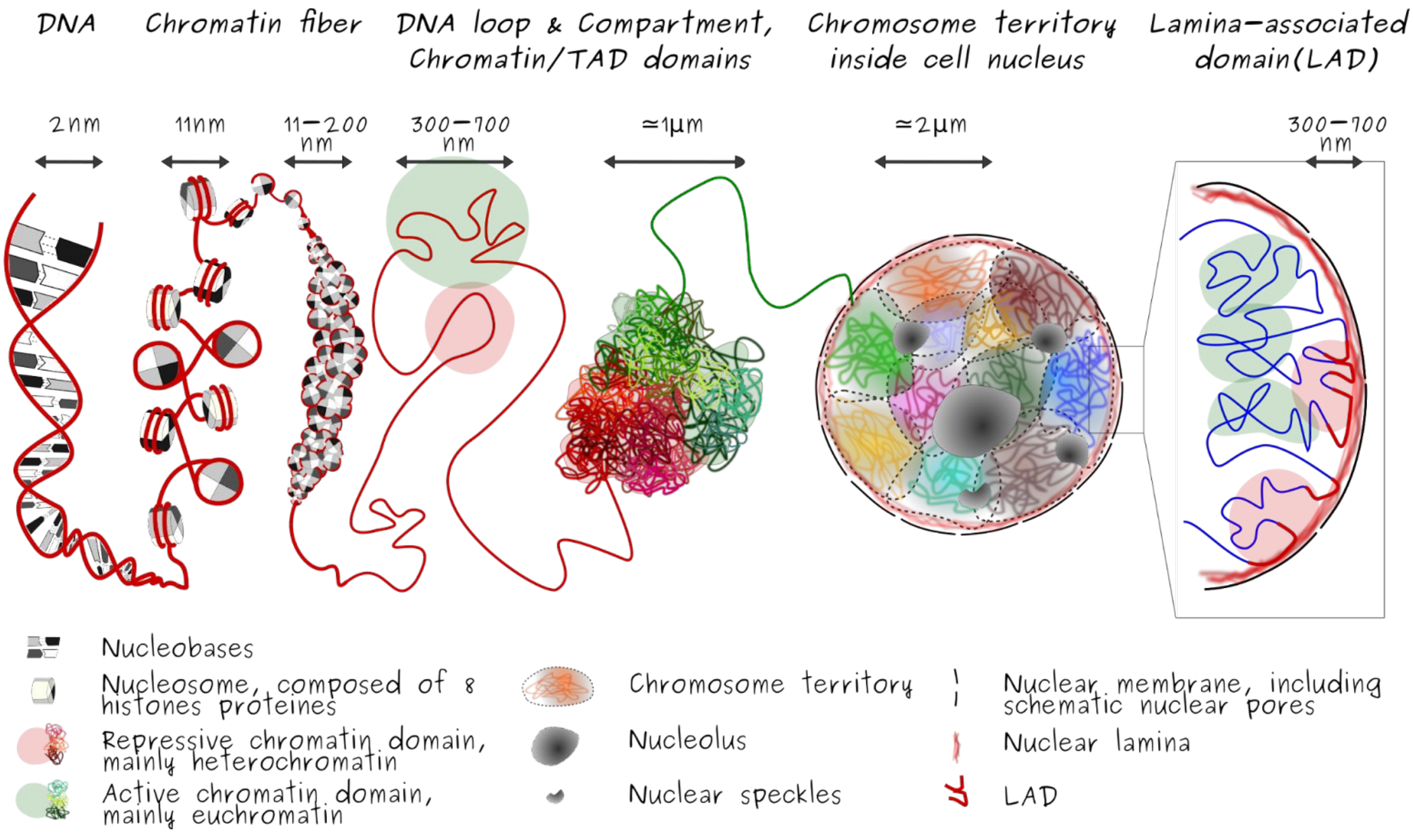
Lysosome



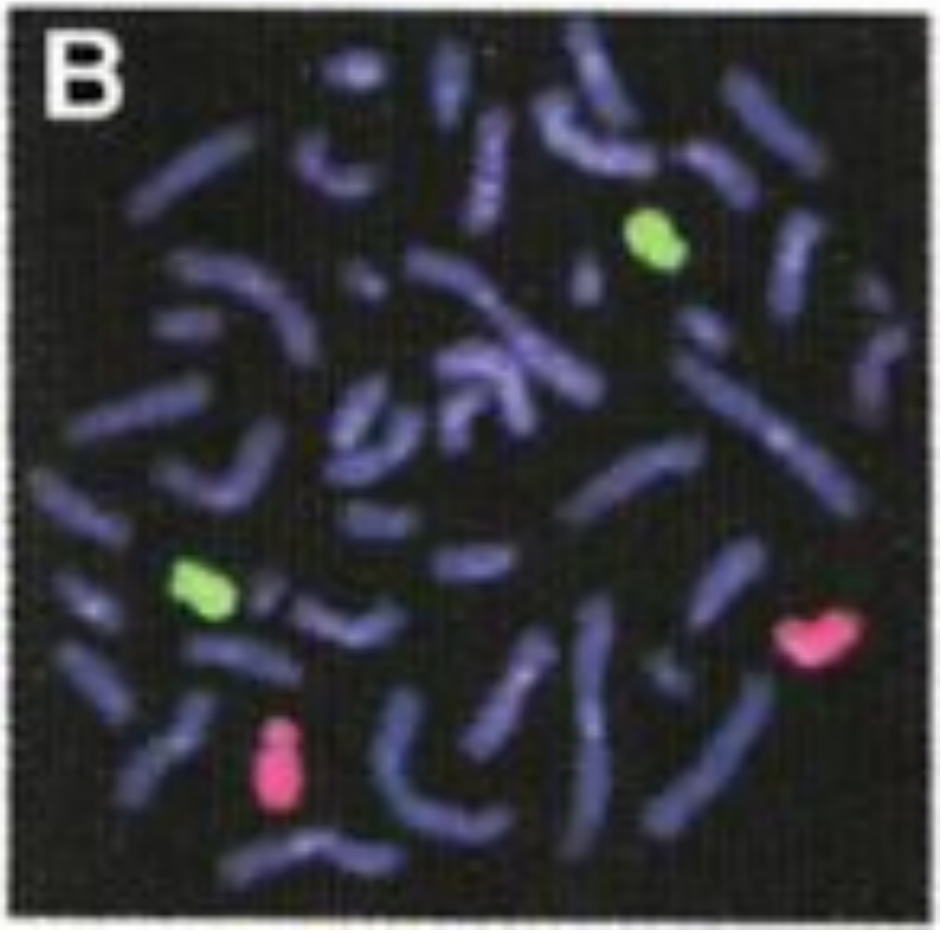
Endoplasmic Reticulum



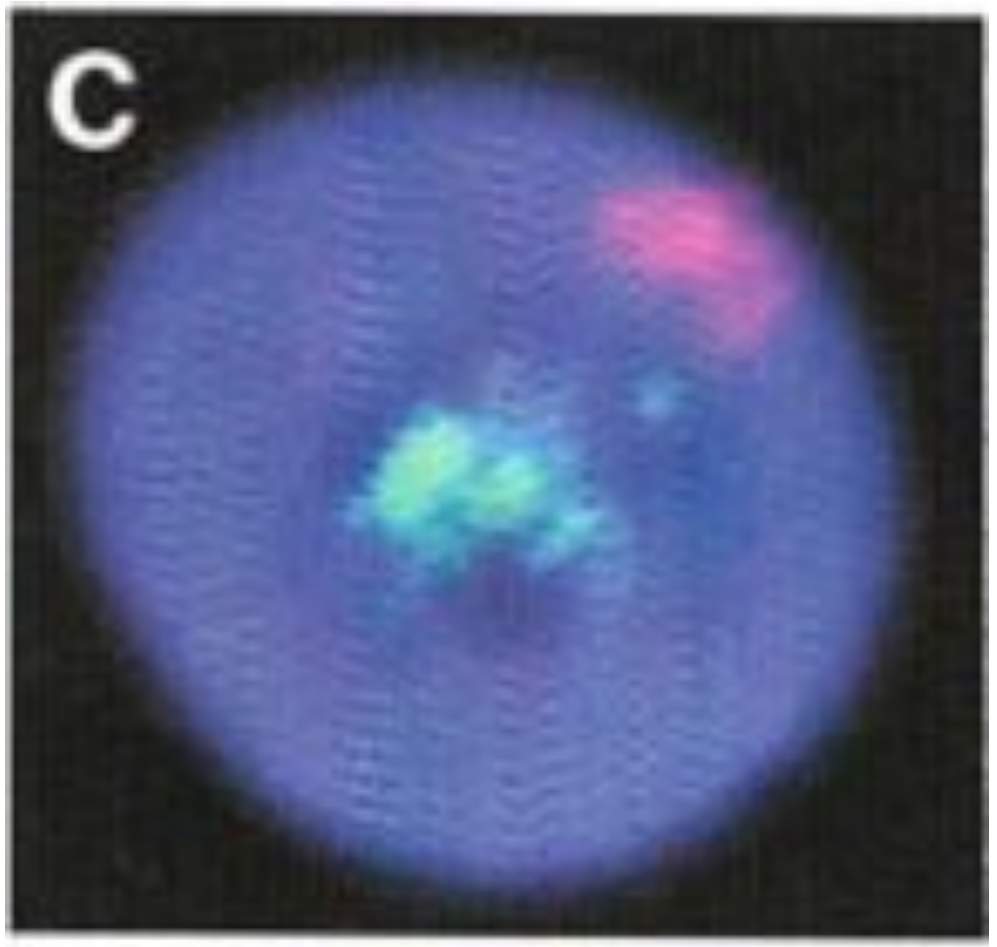
Created with BioRender


DNA
Chromatin fiber
DNA loop & Compartment, Chromatin/TAD domains
Chromosome territory inside cell nucleus
Lamina-associated domain(LAD)
2nm
11nm
11–200 nm
300–700 nm
≃1µm
≃2µm
300–700 nm
Nucleobases
Nucleosome, composed of 8 histones proteines
Repressive chromatin domain, mainly heterochromatin
Active chromatin domain, mainly euchromatin
Chromosome territory
Nucleolus
Nuclear speckles
Nuclear membrane, including schematic nuclear pores
Nuclear lamina
LAD

Direct visualization of CTs using Fluorescence In Situ Hybridization (FISH): 1980 onwards

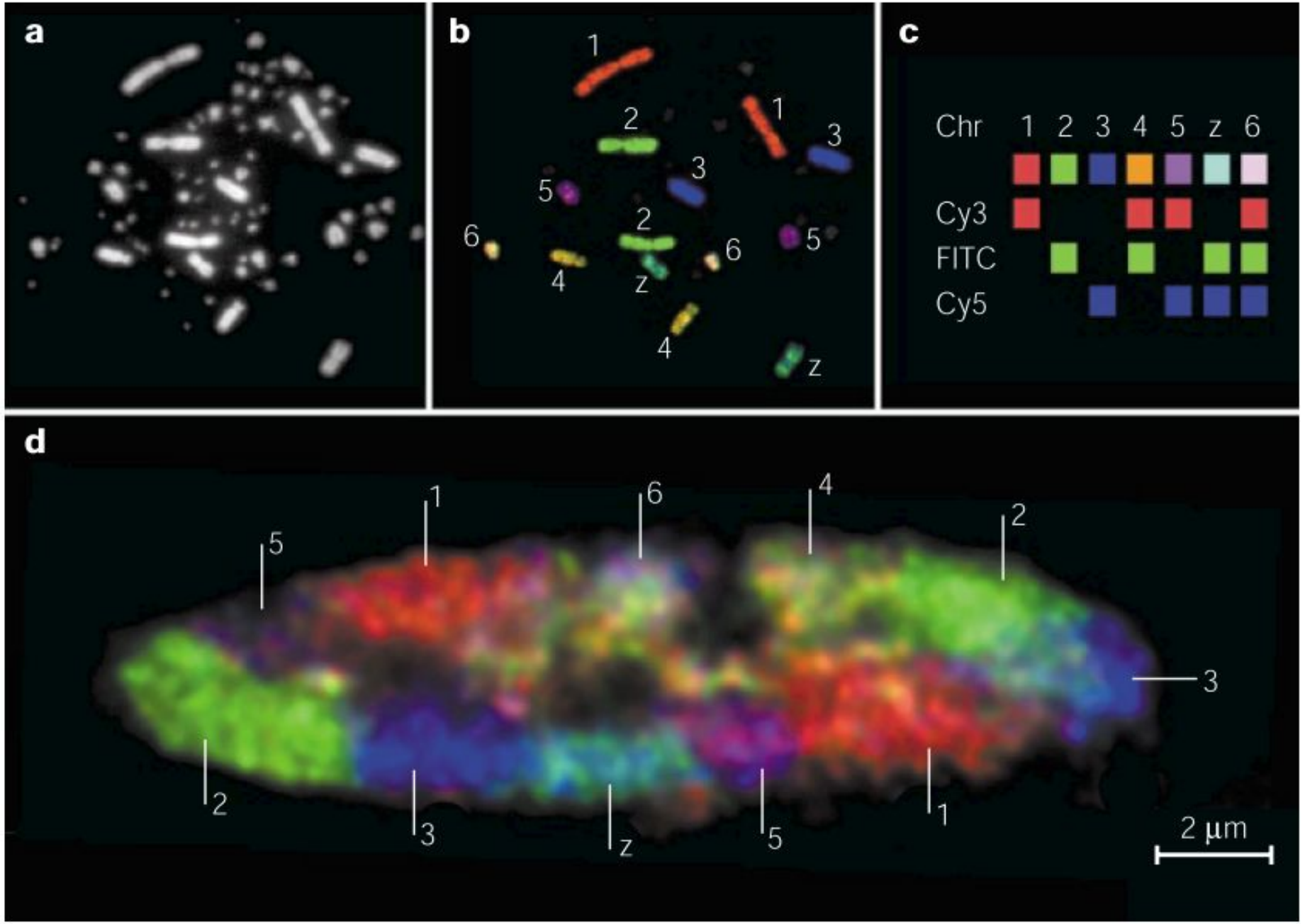


Chr 18 in red, Chr 19 in green by FISH in a human metaphase plate



Nucleus of human lymphoblastoid cell after 3D FISH, Chr 18 in red, Chr 19 in green

Chromosome territories in the chicken cell nucleus visualized by chromosome paint



Cremer and Cremer 2001